Toward Efficient Multimodal Recommendation: From Content Understanding to Personalized Generation

MA Yunshan (马云山)
EReL@MIR Workshop @ ACM Multimedia 2026

A Short Bio:

- 2025.1-now: Assistant Professor, School of Computing and Information Systems, **Singapore Management University**
- 2022.4-2024.12: Postdoctoral Research Fellow in NExT++ Research Center & NCL, NUS
- 2017.8-2022.3: PhD Candidate, School of Computing, NUS, supervised by Prof. Chua Tat-Seng
- Research Interest: Multimodal Recommendation, Bundle Generation, Multimodal Event Forecasting
- Homepage: https://mysbupt.github.io/

© Copyright Singapore Management University. All Rights Reserved.

Outline: Efficient Multimodal Recommendation

Part	Our works	Efficiency lever
I **Relational content representation**	DFF (ICMR 2026, Best Paper Award)	Light fine-tuning (LoRA) → frozen backbones, late fusion
II **Compose items**	DDBC (ICLR 2026)	Generate many items at once with discrete diffusion
III **Generate content**	DualFashion R2G-Map (ongoing work)	Offload rollouts from image to text; efficient backbone

Guiding question: how can recommenders benefit from large multimodal models efficiently?

© Copyright Singapore Management University. All Rights Reserved.

PART I

Relational Content Representation

Foundation models understand content well, but they are slow

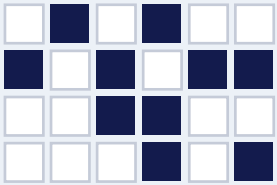
Fine-tune: CIRP (MM 2024) · LARP (KDD 2024) · TPAD (TOIS 2026)
Freeze: DFF (ICMR 2026, Best Paper Award)

© Copyright Singapore Management University. All Rights Reserved.

Two Kinds of Data in Recommender Systems

Behavior data

User–item interactions



Sequences

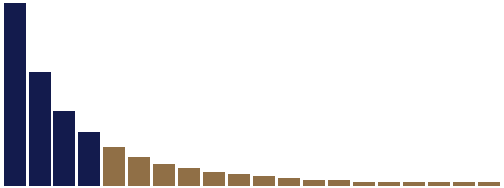


Collaborative filtering and sequential patterns: who interacts with what, and in which order

&

Item content data

- **Text** — titles, descriptions: semantics, attributes
- **Image** — product photos, covers: style and appearance
- **Audio** — music and speech: mood, genre, rhythm
- **Video** — micro-videos: motion, storyline, editing



head → long tail and cold-start items

- Many recommenders **over-rely on behavior data** (CF, sequential patterns)
- Long-tail and cold-start items have few interactions: **content is the main signal**
- Content understanding now runs on **large foundation models, which are slow**

© Copyright Singapore Management University. All Rights Reserved.

CIRP: Relation-Aware Item Representations

CIRP: cross-item relational pre-training for multimodal product bundling

Motivation:

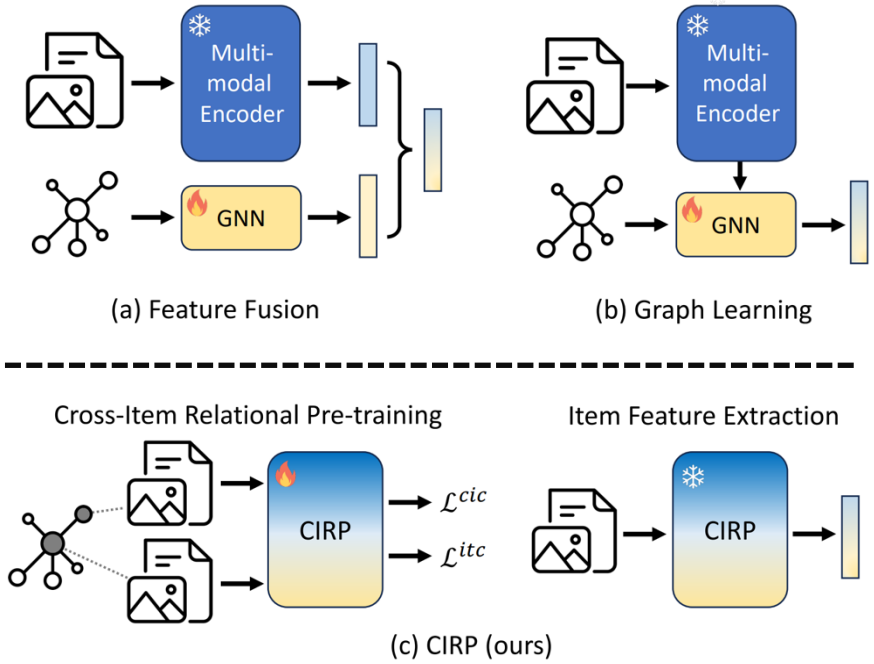
Explicitly distill cross-item relations into VLMs.
- Cross-Item Relational Pre-training (**CIRP**)

Pre-training Stage:

- Image-text contrastive (ITC) loss retains multimodal understanding.
- Cross-item contrastive (CIC) loss captures cross-item relations.

Inference Stage:

- Generate relation-aware multimodal representations, even for cold-start items.



CIRP: Cross-Item Relational Pre-training for Multimodal Product Bundling. Yunshan Ma et al. ACM MM 2024.

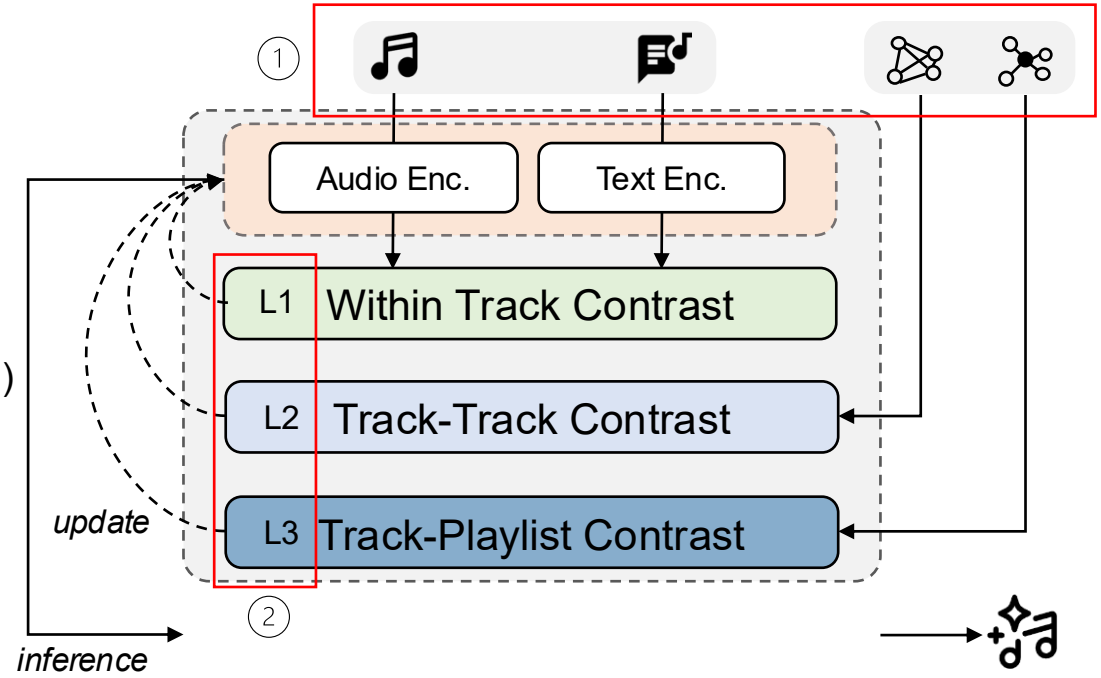
© Copyright Singapore Management University. All Rights Reserved.

LARP: Audio–Text Relational Pre-Training

LARP: language audio relational pre-training for cold-start playlist continuation

① Three ***modalities*** of data (introduce audio):
- Audio - **new**
- Language
- Relational

② Extend CIRP from two losses (ITC and CIC) to three ***stages*** of contrastive learning:
- Within Track (WTC) - ITC
- Track to Track (TTC) - CIC
- Track to Playlist (TPC) - **new**



LARP: Language Audio Relational Pre-training for Cold-Start Playlist Continuation. Rebecca Salganik et al. KDD 2024.

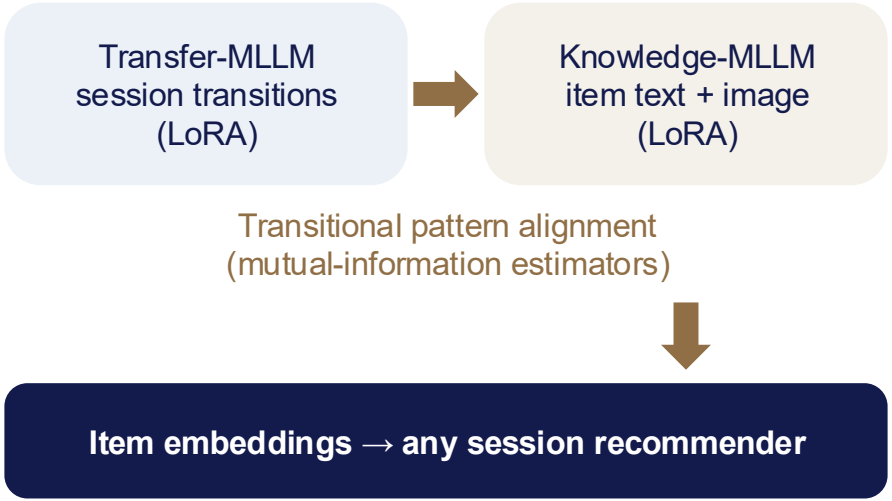
© Copyright Singapore Management University. All Rights Reserved.

TPAD: Distilling Transitions into an MLLM

Motivation

- Session recommendation: predict the next item
- Multimodal LLMs know what an item is…
- …but not which item tends to follow which
- Distill transitional patterns into the MLLM
- LoRA on a frozen 7B backbone

EFFICIENCY



Efficiency: LoRA on a frozen Llama-2-7B; distilled embeddings plug into existing recommenders.

Su, Liu, Zhong, Chen, Ma, et al. Distilling Transitional Pattern to Large Language Models for Multimodal Session-based Recommendation. ACM TOIS 2026.

© Copyright Singapore Management University. All Rights Reserved.

Two Paradigms: Fine-Tune or Freeze

EFFICIENCY

Goal: use behavior and relational data to make better use of content

PARADIGM A · FINE-TUNE

Inject behavior into the backbone

- Fine-tune the content backbone on relations
- Co-occurrence: CIRP and LARP
- Session transitions: TPAD (LoRA)
- Output: relation-aware item features

Cost: training passes through a large backbone

PARADIGM B · FREEZE

Understand content, fuse late

- Frozen backbone: content understanding only
- Behavior: a separate recommender
- Late fusion of the two: DFF
- 25× faster than end-to-end fine-tuning

Cost: one forward pass per item, offline

Increasingly efficient: full training → light fine-tuning → frozen backbone + late fusion

© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content

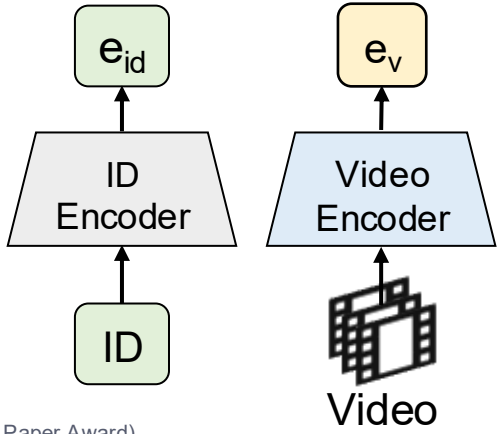


Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content



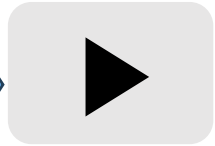


Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

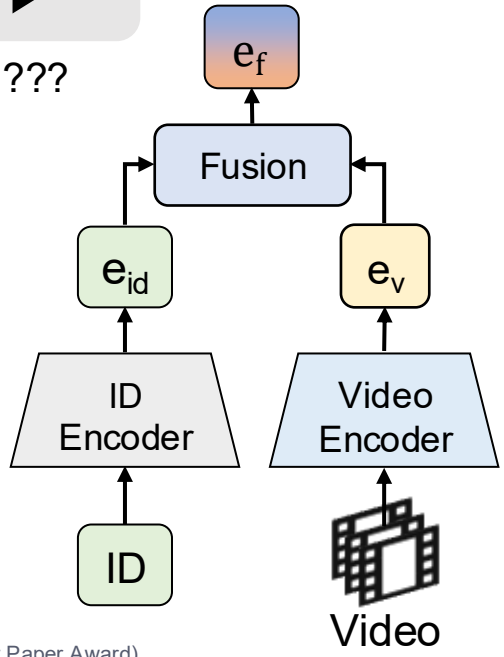
© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content





???



IDRec e_{id}^4 e_{id}^{16} e_{id}^{28} ... e_{id}^t

ContentRec e_v^4 e_v^{16} e_v^{28} ... e_v^t

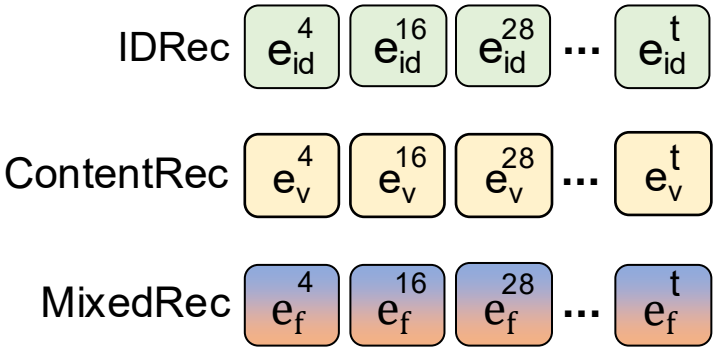
Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

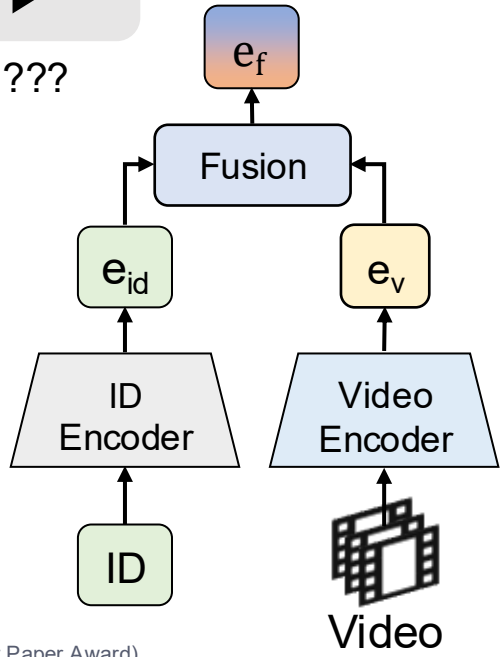
© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content



???

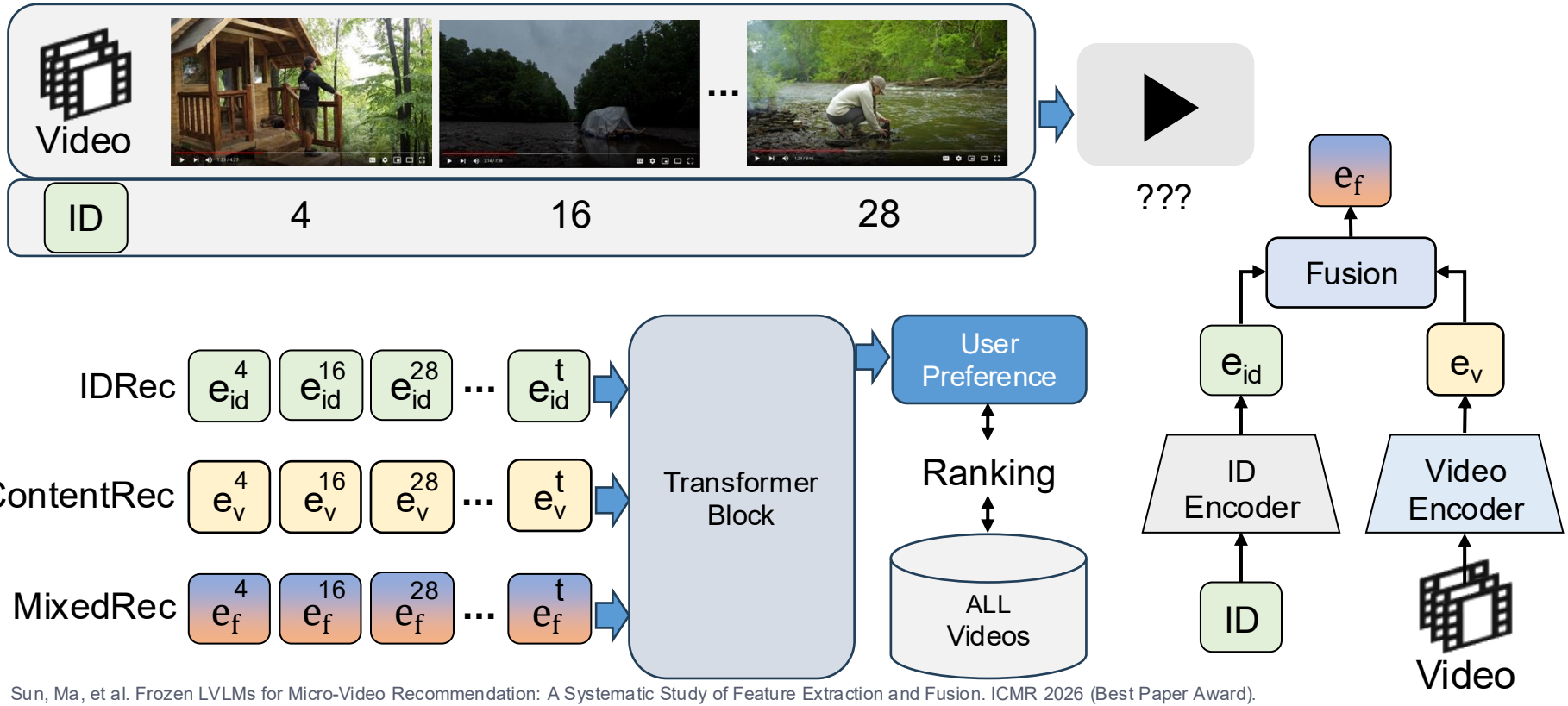




Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content



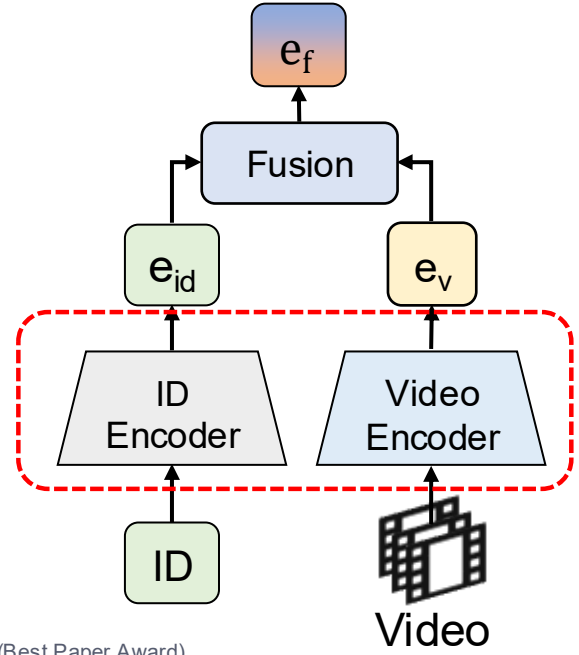
Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Micro-Video Rec with ID and Content

Question:

What are more effective feature extraction strategies for micro-video recommendation?



Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

Video Encoders: From CNNs to Transformers

From frame-level image feaature to spatio-temporal video representations

Image Encoder Evolution

2012 — **AlexNet**
Deep CNN

2015 — **ResNet**
Residual Learing

2020 — **ViT**
Vision transformer

Video Encoder Evolution

Improved Dense Trajectories(2012)
Feature based on dense trajectories

Two-stream ConvNet(2014)
RGB stream + Temporal stream

C3D (2015) / I3D (2017) / R3D(2018)
3D Convolutional Network

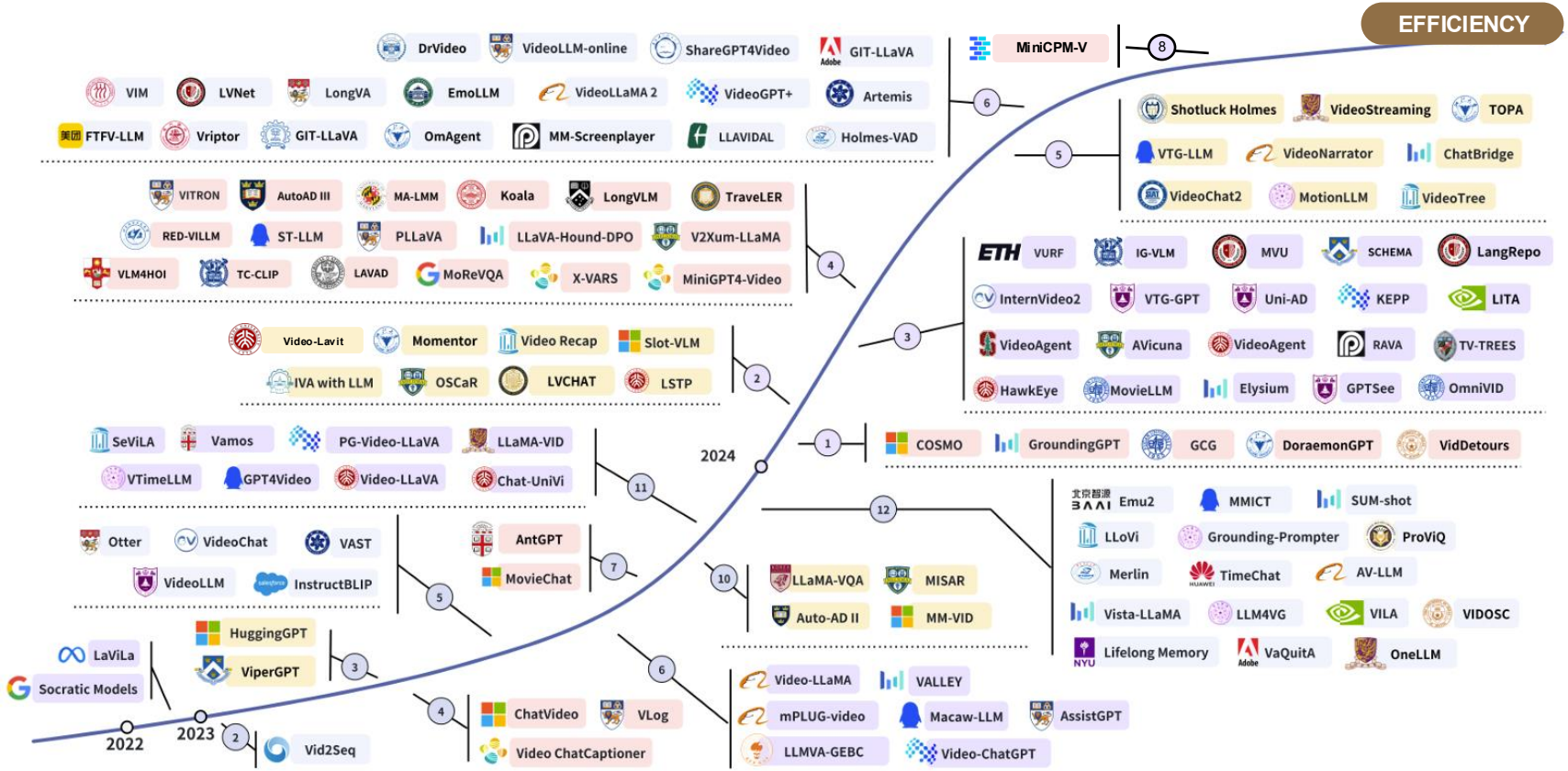
SlowFast(2019)
Dual-pathway network

MViT(2021)
Multiscale Vision Transformer

VideoMAE(2022)
Masked Autoencoder

© Copyright Singapore Management University. All Rights Reserved.

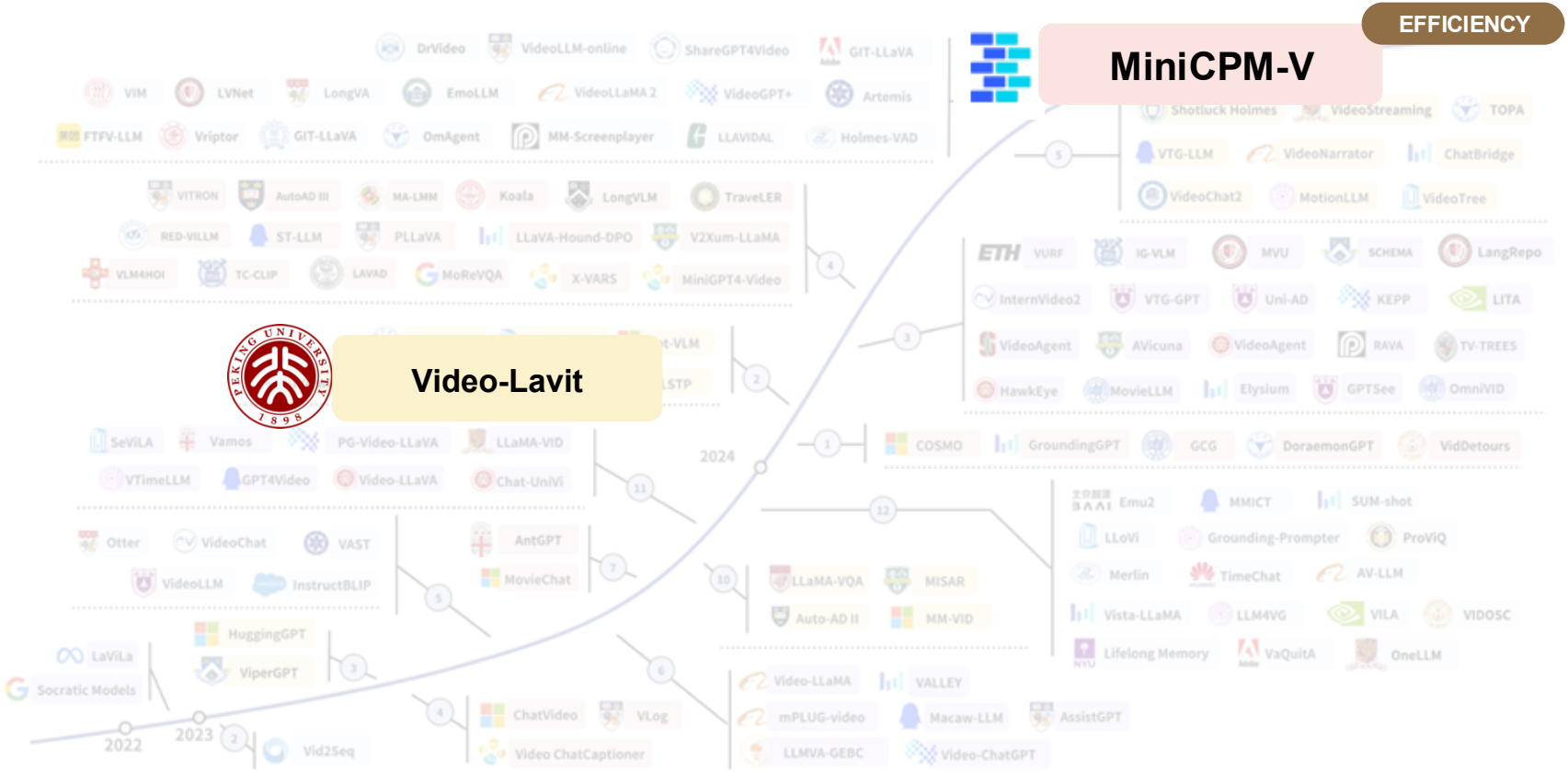
Content Encoder: Large Video-Language Models



[1] Y. Tang et al., "Video Understanding With Large Language Models: A Survey," in IEEE Transactions on Circuits and Systems for Video Technology, vol. 36, no. 2, pp. 1355-1376, Feb. 2026, doi: 10.1109/TCSVT.2025.3566695.

© Copyright Singapore Management University. All Rights Reserved.

Content Encoder: Large Video-Language Models



© Copyright Singapore Management University. All Rights Reserved.

Content Encoder: Large Video-Language Models

EFFICIENCY

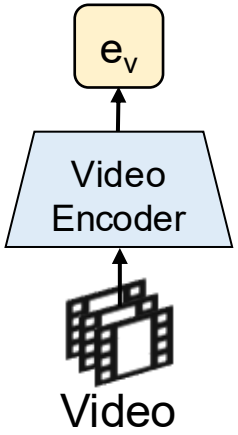
Model	Scale	Runtime on per video	Video modeling idea
Video-LaVIT	7B	8.475s	Keyframes + motion cues; compact tokens
MiniCPM-V 2.6	8B	4.423s	Video as sampled image sequence
MiniCPM-V 4.5	9B	3.523s	3D Resampler for spatio-temporal representation

- Key criterion:
  - open-source
  - compact yet capable models
  - support single-GPU inference

[1] Jin Y, Sun Z, Xu K, et al. Video-LaVIT: Unified video-language pre-training with decoupled visual-motional tokenization. ICML, 2024.
[2] Yao Y, Yu T, Zhang A, et al. MiniCPM-V: A GPT-4V level MLLM on your phone. arXiv, 2024.
[3] Yu T, Wang Z, Wang C, et al. MiniCPM-V 4.5: Cooking efficient MLLMs via architecture, data, and training recipe. arXiv, 2025.

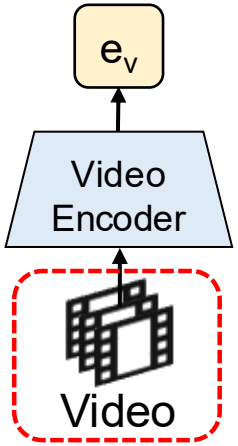
© Copyright Singapore Management University. All Rights Reserved.

LVLM Features: Captions or Hidden States?



© Copyright Singapore Management University. All Rights Reserved.

LVLM Features: Captions or Hidden States?





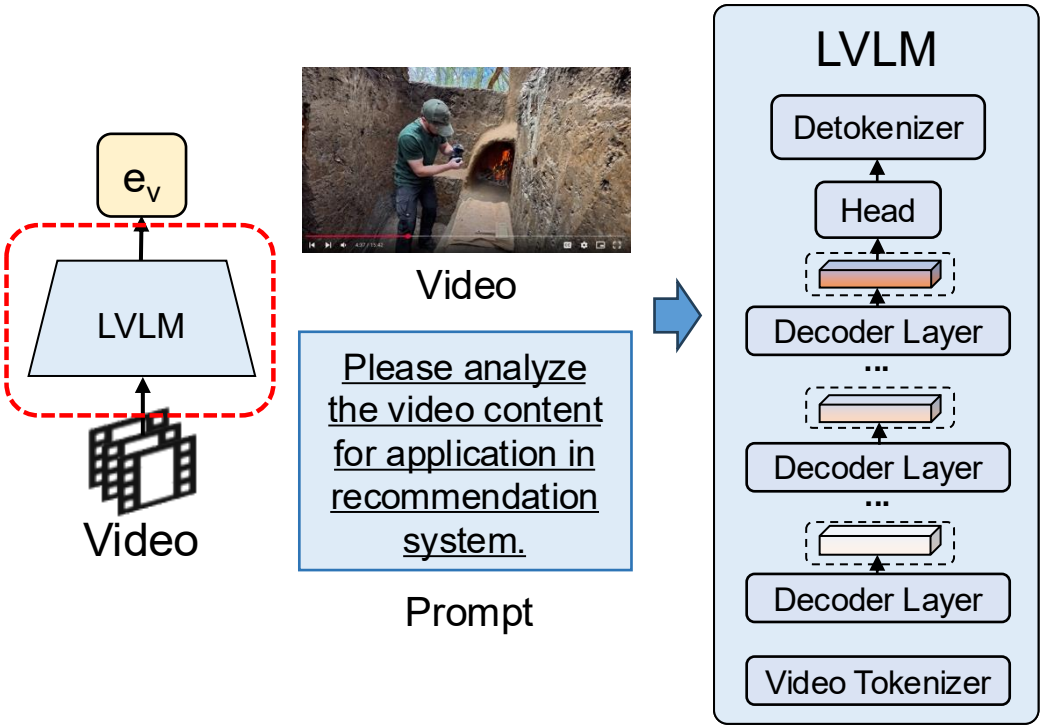
Video

Please analyze the video content for application in recommendation system.

Prompt

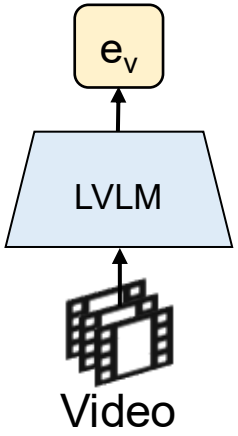
© Copyright Singapore Management University. All Rights Reserved.

LVLM Features: Captions or Hidden States?



© Copyright Singapore Management University. All Rights Reserved.

LVLM Features: Captions or Hidden States?

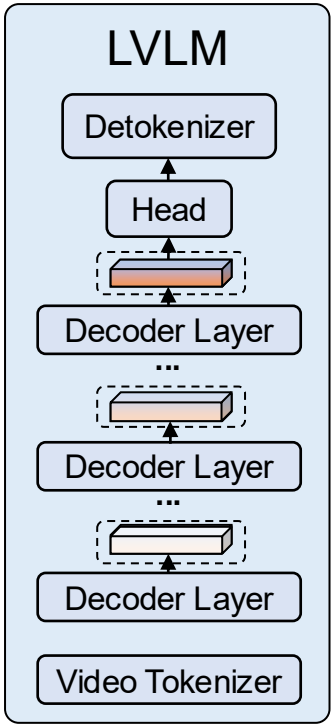




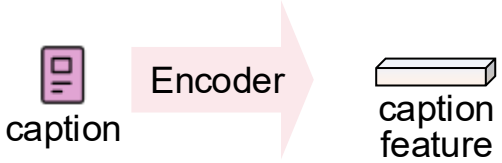
Video

Please analyze the video content for application in recommendation system.

Prompt

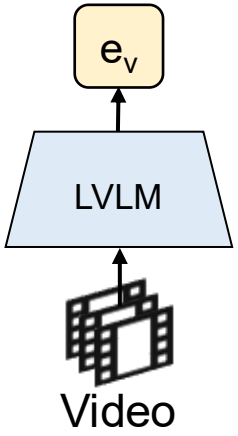


➢ **Caption-Centric(CC)**



© Copyright Singapore Management University. All Rights Reserved.

LVLM Features: Captions or Hidden States?

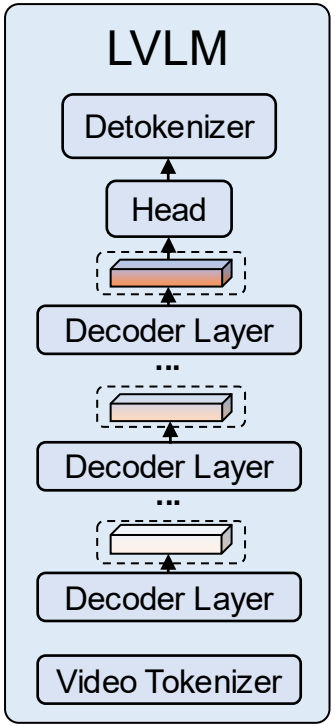




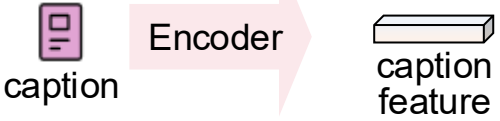
Video

Please analyze the video content for application in recommendation system.

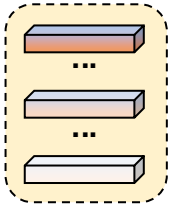
Prompt



➢ **Caption-Centric(CC)**



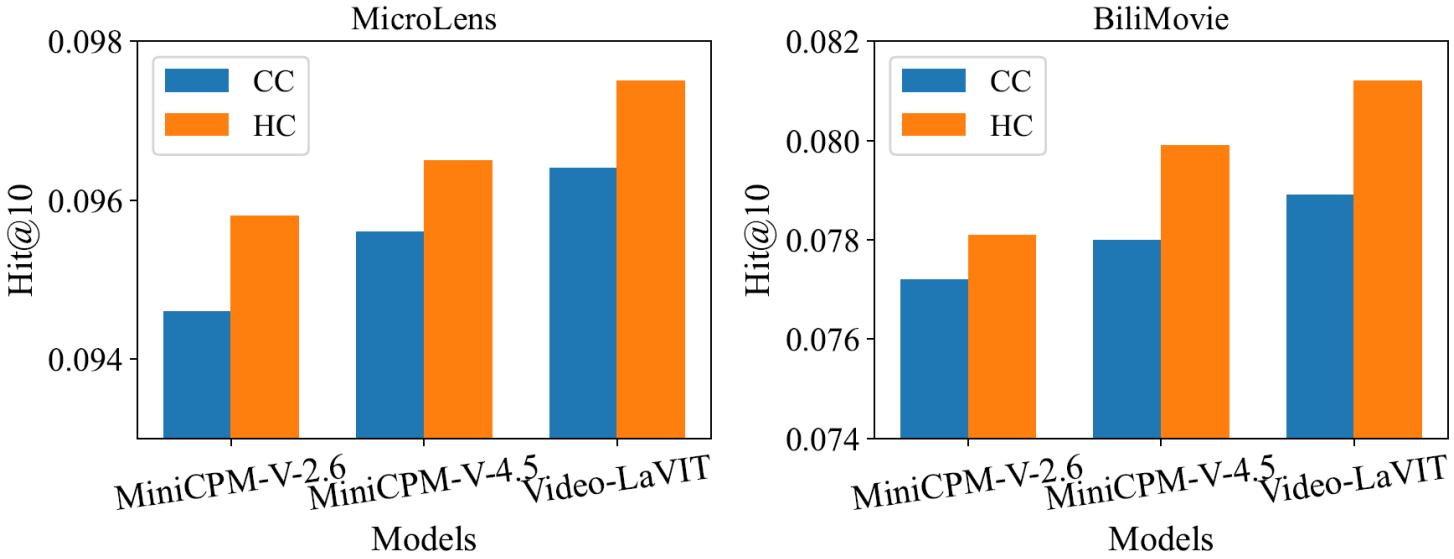
➢ **Hidden-state-Centric(HC)**



LVLM hidden states

© Copyright Singapore Management University. All Rights Reserved.

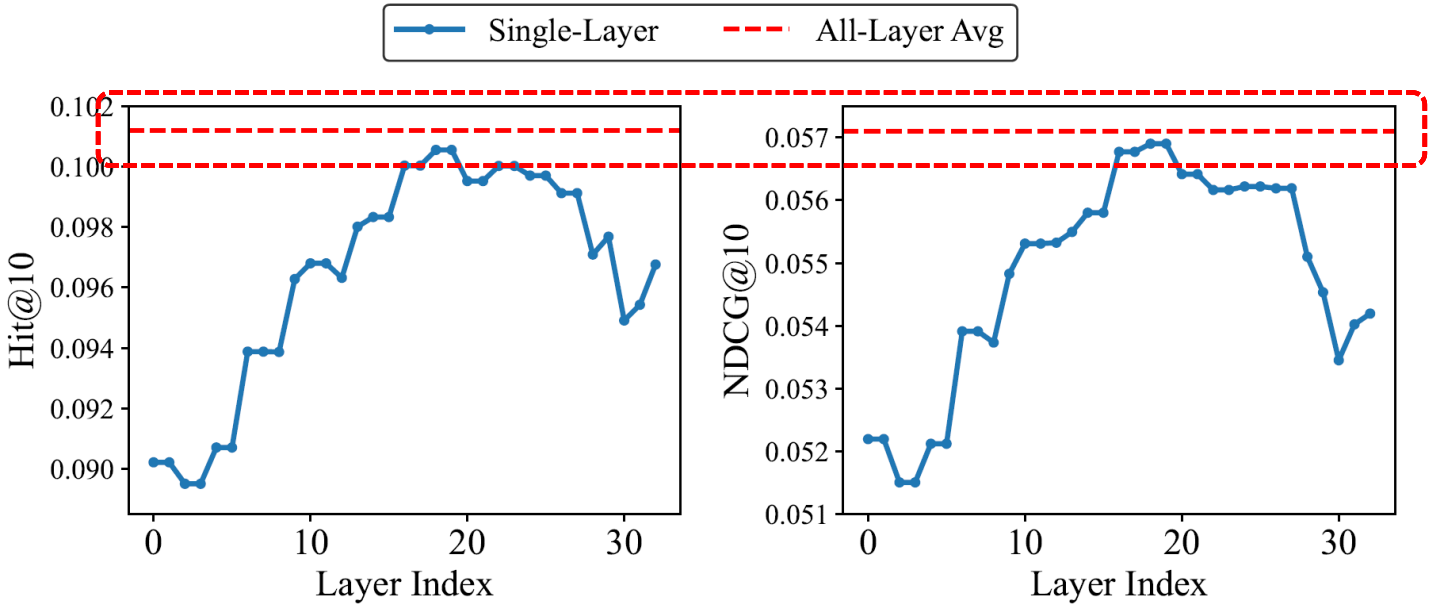
Hidden States Beat Captions



- HC > CC across all settings.
- Hidden states preserve richer visual semantics than captions.
- Directly using decoder states is more effective for MVR.

© Copyright Singapore Management University. All Rights Reserved.

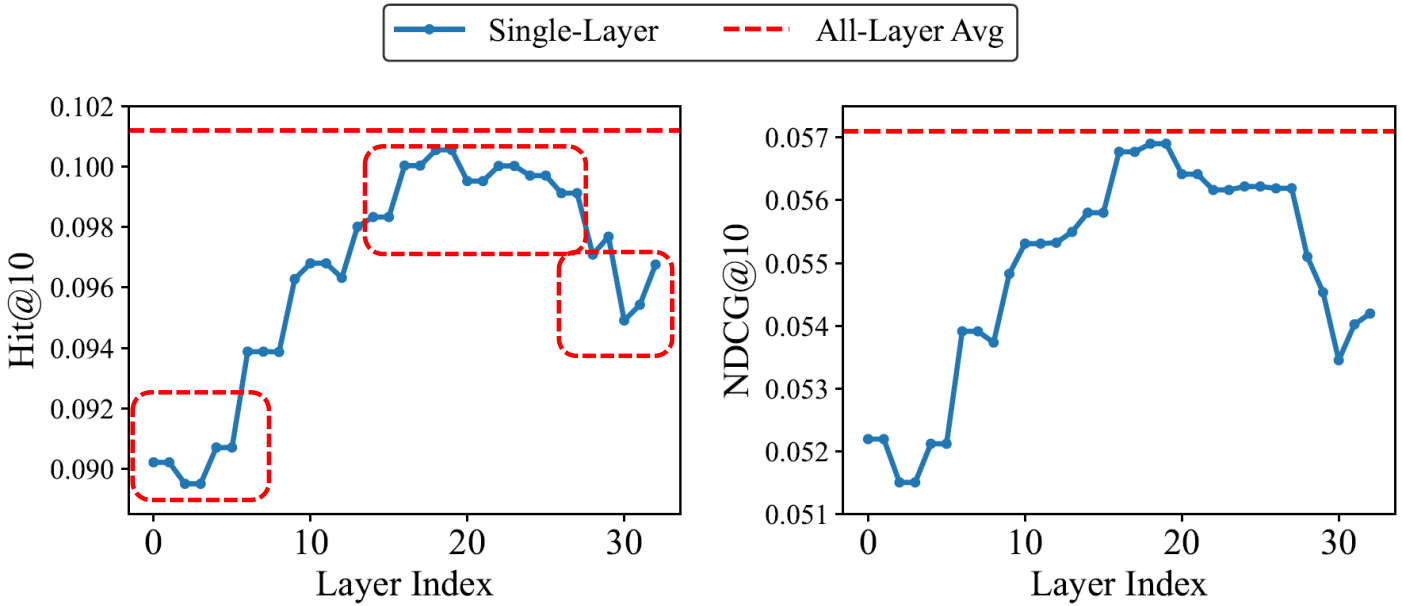
Which Layer? All-Layer Fusion Wins



- **All-layer Avg performs best.**

© Copyright Singapore Management University. All Rights Reserved.

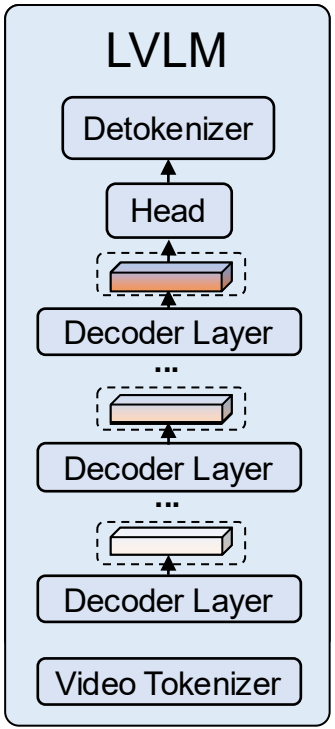
Which Layer? All-Layer Fusion Wins

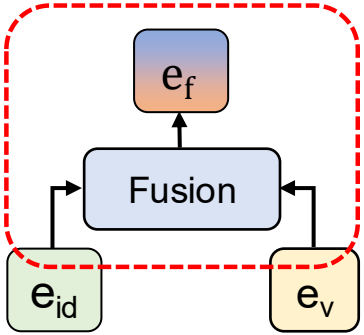


- **All-layer Avg performs best.**
- **Middle layers are more informative.**
- **Multi-layer fusion captures complementary semantics.**

© Copyright Singapore Management University. All Rights Reserved.

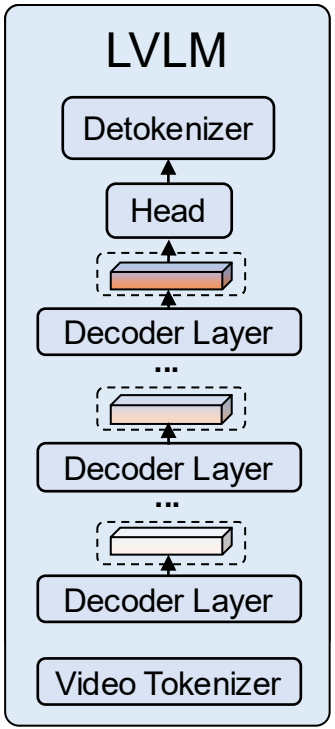
DFF: Multi-Layer Fusion + Gate Fusion



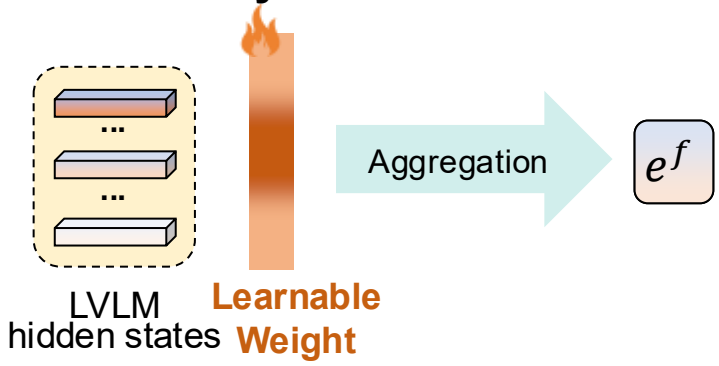


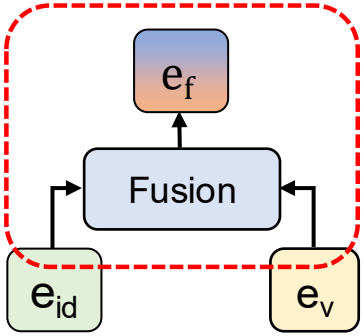
© Copyright Singapore Management University. All Rights Reserved.

DFF: Multi-Layer Fusion + Gate Fusion



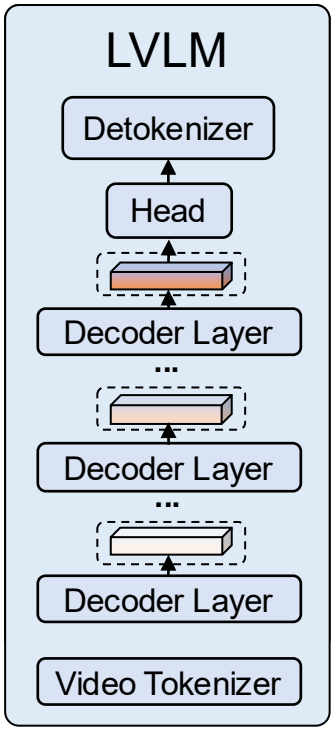
➢ **Multi-Layer Fusion**



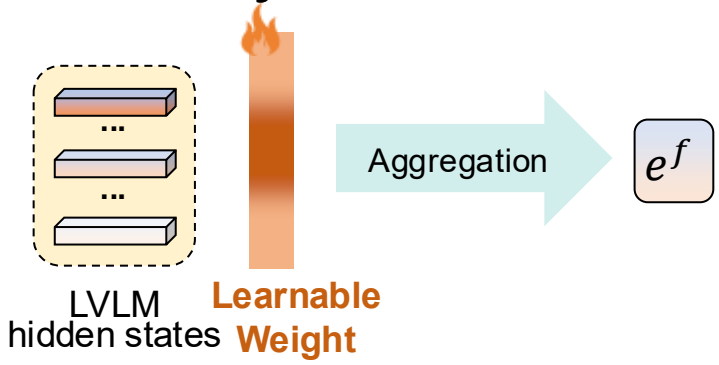


© Copyright Singapore Management University. All Rights Reserved.

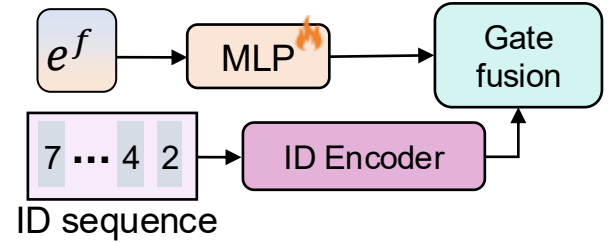
DFF: Multi-Layer Fusion + Gate Fusion

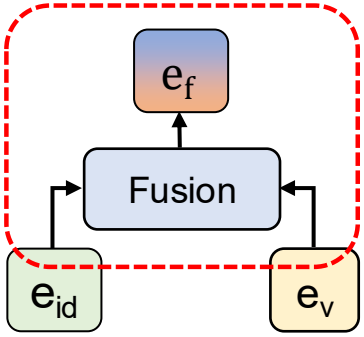


➢ Multi-Layer Fusion



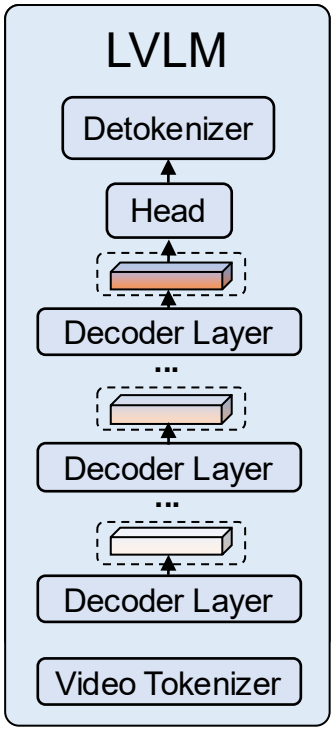
➢ Gate Fusion



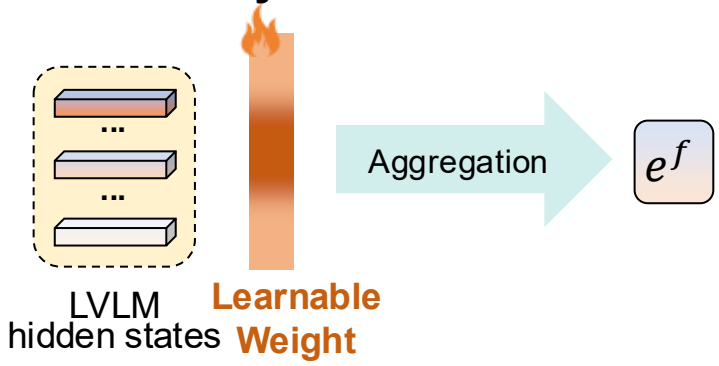


© Copyright Singapore Management University. All Rights Reserved.

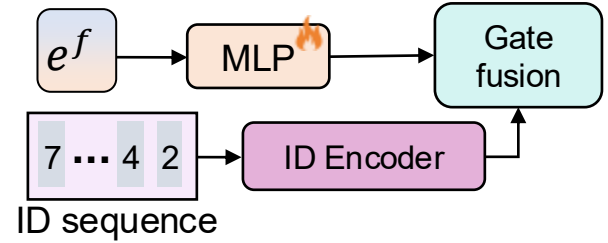
DFF: Multi-Layer Fusion + Gate Fusion

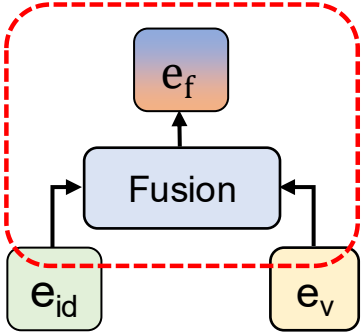


➢ Multi-Layer Fusion



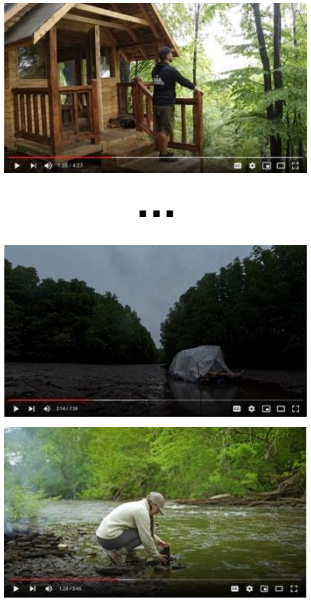
➢ Gate Fusion

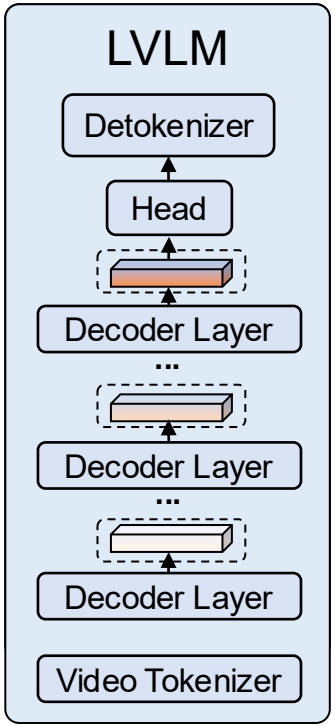




© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs





© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs

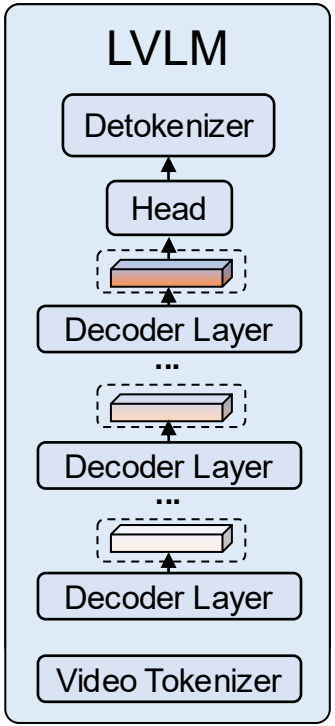


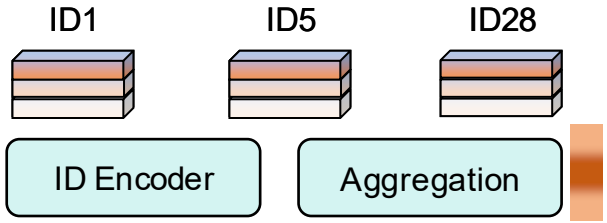
...





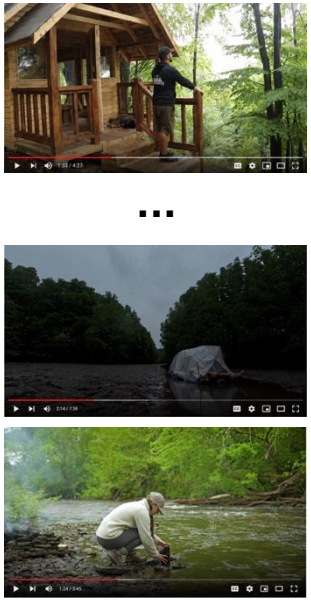
Video View history

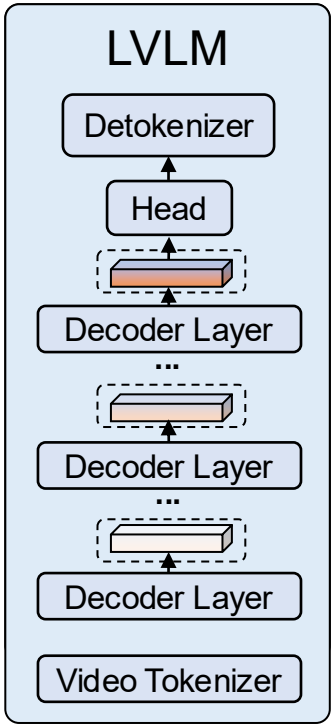


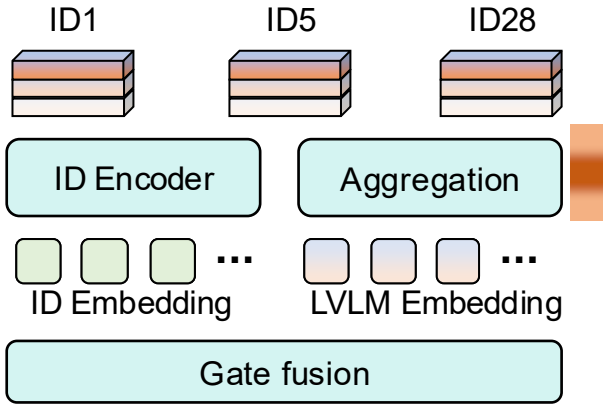


© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs

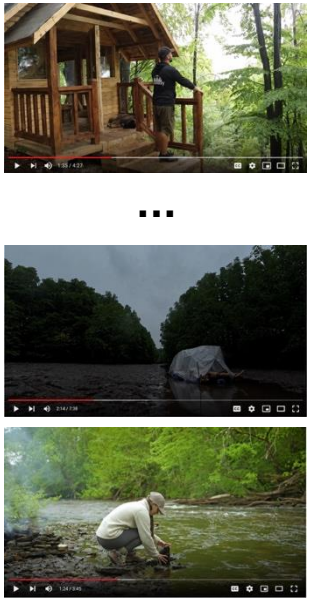


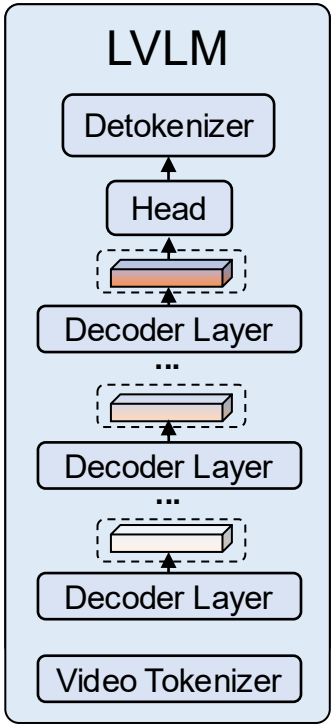


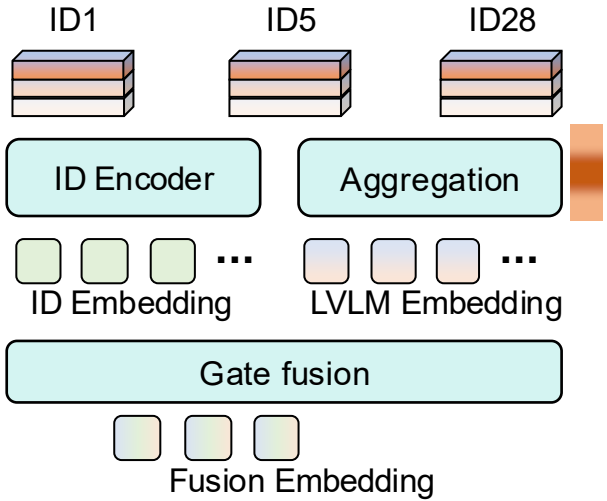


© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs

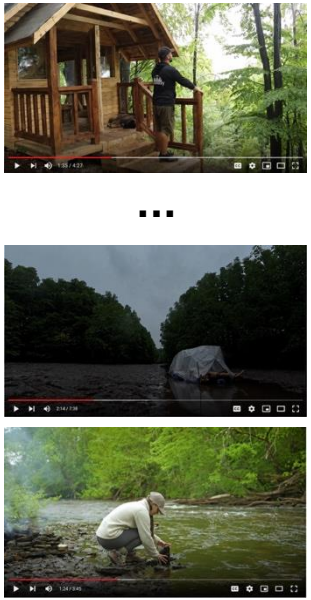




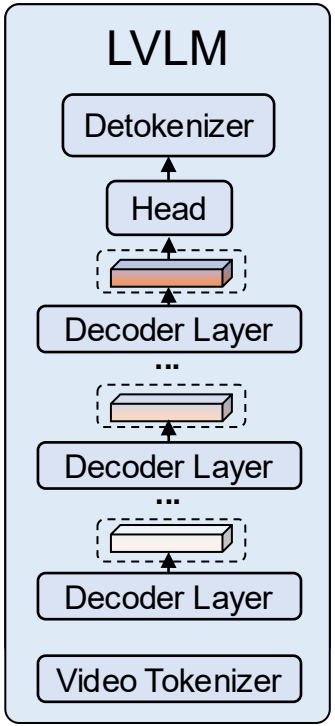


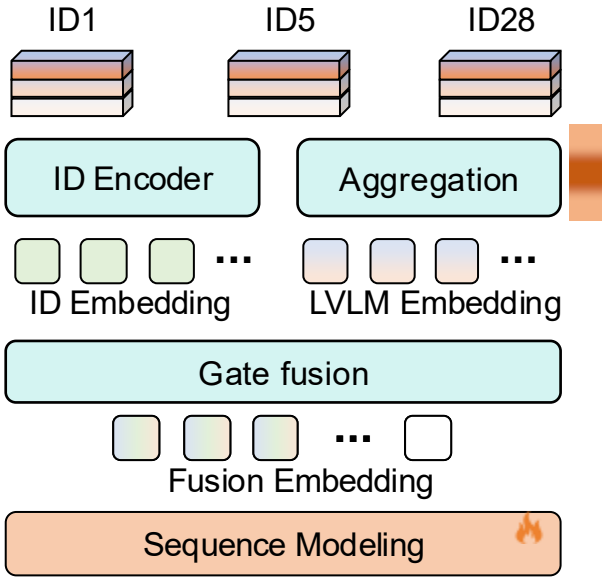
© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs



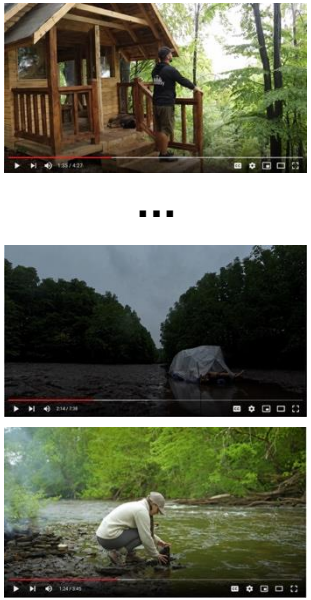
Video View history

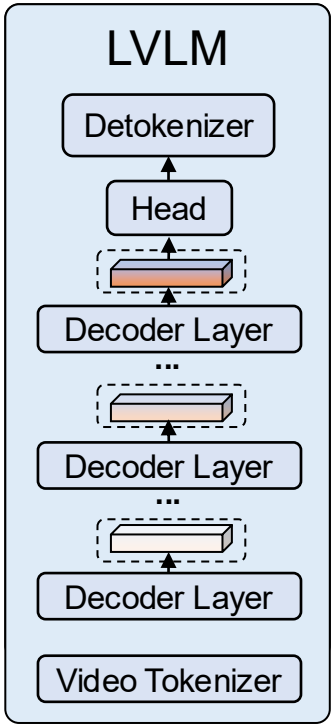


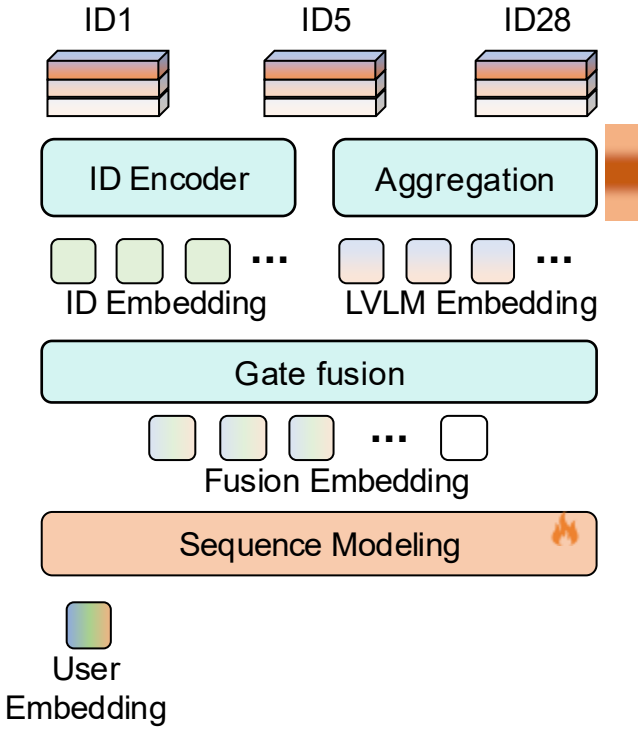


© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs

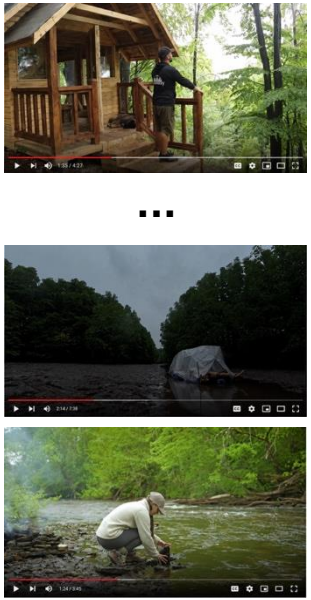


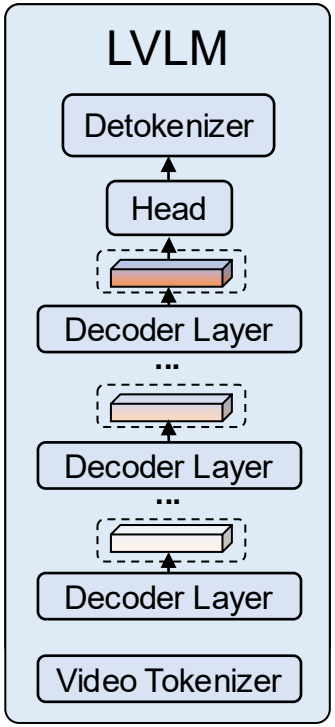


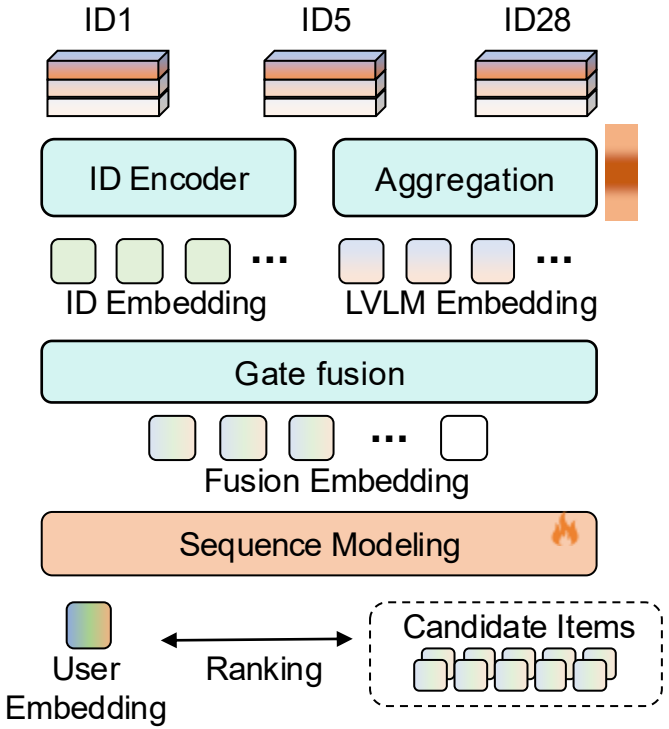


© Copyright Singapore Management University. All Rights Reserved.

DFF: Dual Feature Fusion on Frozen LVLMs







© Copyright Singapore Management University. All Rights Reserved.

DFF: Overall Performance

Model	MicroLens				Bili_movie			
	H@10	N@10	H@20	N@20	H@10	N@10	H@20	N@20
IDRec (official baselines from MicroLens)								
SASRec	0.0909	0.0517	0.1278	0.0610	0.0760	0.0441	0.1074	0.0520
GRU4Rec	0.0782	0.0423	0.1147	0.0515	0.0621	0.0427	0.1012	0.0462
NextItNet	0.0805	0.0442	0.1175	0.0535	0.0691	0.0433	0.1056	0.0498
VideoRec (official baselines from MicroLens)								
SASRec	0.0948	0.0515	0.1364	0.0619	0.0769	0.0449	0.1102	0.0523
GRU4Rec	0.0954	0.0517	0.1377	0.0623	0.0770	0.0451	0.1132	0.0527
NextItNet	0.0862	0.0466	0.1246	0.0562	0.0712	0.436	0.1071	0.0506
caption-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0946	0.0533	0.1317	0.0627	0.0772	0.0449	0.1109	0.0513
MiniCPM-V 4.5	0.0956	0.0537	0.1333	0.0632	0.0780	0.0453	0.1130	0.0524
Video-LaVIT	0.0964	0.0541	0.1334	0.0634	0.0789	0.0450	0.1144	0.0533
hidden-state-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0958	0.0540	0.1341	0.0636	0.0781	0.0452	0.1139	0.0522
MiniCPM-V 4.5	0.0965	0.0542	0.1343	0.0637	0.0799	0.0449	0.1142	0.0531
Video-LaVIT	0.0975	0.0545	0.1353	0.0640	0.0812	0.0454	0.1151	0.0539
Our Method								
DFF *(ours)*	**0.1020***	**0.0577***	**0.1414***	**0.0676***	**0.0839***	**0.0471***	**0.1156***	**0.0551***
Imp.	4.61%	5.87%	4.50%	5.62%	3.32%	3.74%	0.43%	2.22%

Results:

- **DFF consistently achieves the best results across all metrics.**

Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Overall Performance

Model	MicroLens				Bili_movie			
	H@10	N@10	H@20	N@20	H@10	N@10	H@20	N@20
IDRec (official baselines from MicroLens)								
SASRec	0.0909	0.0517	0.1278	0.0610	0.0760	0.0441	0.1074	0.0520
GRU4Rec	0.0782	0.0423	0.1147	0.0515	0.0621	0.0427	0.1012	0.0462
NextItNet	0.0805	0.0442	0.1175	0.0535	0.0691	0.0433	0.1056	0.0498
VideoRec (official baselines from MicroLens)								
SASRec	0.0948	0.0515	0.1364	0.0619	0.0769	0.0449	0.1102	0.0523
GRU4Rec	0.0954	0.0517	0.1377	0.0623	0.0770	0.0451	0.1132	0.0527
NextItNet	0.0862	0.0466	0.1246	0.0562	0.0712	0.436	0.1071	0.0506
caption-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0946	0.0533	0.1317	0.0627	0.0772	0.0449	0.1109	0.0513
MiniCPM-V 4.5	0.0956	0.0537	0.1333	0.0632	0.0780	0.0453	0.1130	0.0524
Video-LaVIT	0.0964	0.0541	0.1334	0.0634	0.0789	0.0450	0.1144	0.0533
hidden-state-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0958	0.0540	0.1341	0.0636	0.0781	0.0452	0.1139	0.0522
MiniCPM-V 4.5	0.0965	0.0542	0.1343	0.0637	0.0799	0.0449	0.1142	0.0531
Video-LaVIT	0.0975	0.0545	0.1353	0.0640	0.0812	0.0454	0.1151	0.0539
Our Method								
DFF *(ours)*	**0.1020***	**0.0577***	**0.1414***	**0.0676***	**0.0839***	**0.0471***	**0.1156***	**0.0551***
Imp.	4.61%	5.87%	4.50%	5.62%	3.32%	3.74%	0.43%	2.22%

Results:

DFF consistently achieves the best results across all metrics.

Multi-layer LVLM feature fusion outperforms both official baselines and single-layer frozen-LVLM variants.

Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Overall Performance

Model	MicroLens				Bili_movie			
	H@10	N@10	H@20	N@20	H@10	N@10	H@20	N@20
IDRec (official baselines from MicroLens)								
SASRec	0.0909	0.0517	0.1278	0.0610	0.0760	0.0441	0.1074	0.0520
GRU4Rec	0.0782	0.0423	0.1147	0.0515	0.0621	0.0427	0.1012	0.0462
NextItNet	0.0805	0.0442	0.1175	0.0535	0.0691	0.0433	0.1056	0.0498
VideoRec (official baselines from MicroLens)								
SASRec	0.0948	0.0515	0.1364	0.0619	0.0769	0.0449	0.1102	0.0523
GRU4Rec	0.0954	0.0517	0.1377	0.0623	0.0770	0.0451	0.1132	0.0527
NextItNet	0.0862	0.0466	0.1246	0.0562	0.0712	0.436	0.1071	0.0506
caption-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0946	0.0533	0.1317	0.0627	0.0772	0.0449	0.1109	0.0513
MiniCPM-V 4.5	0.0956	0.0537	0.1333	0.0632	0.0780	0.0453	0.1130	0.0524
Video-LaVIT	0.0964	0.0541	0.1334	0.0634	0.0789	0.0450	0.1144	0.0533
hidden-state-centric feature extraction and ID embedding fusion								
MiniCPM-V 2.6	0.0958	0.0540	0.1341	0.0636	0.0781	0.0452	0.1139	0.0522
MiniCPM-V 4.5	0.0965	0.0542	0.1343	0.0637	0.0799	0.0449	0.1142	0.0531
Video-LaVIT	0.0975	0.0545	0.1353	0.0640	0.0812	0.0454	0.1151	0.0539
Our Method								
DFF *(ours)*	**0.1020***	**0.0577***	**0.1414***	**0.0676***	**0.0839***	**0.0471***	**0.1156***	**0.0551***
Imp.	4.61%	5.87%	4.50%	5.62%	3.32%	3.74%	0.43%	2.22%

Results:

DFF consistently achieves the best results across all metrics.

Multi-layer LVLM feature fusion outperforms both official baselines and single-layer frozen-LVLM variants.

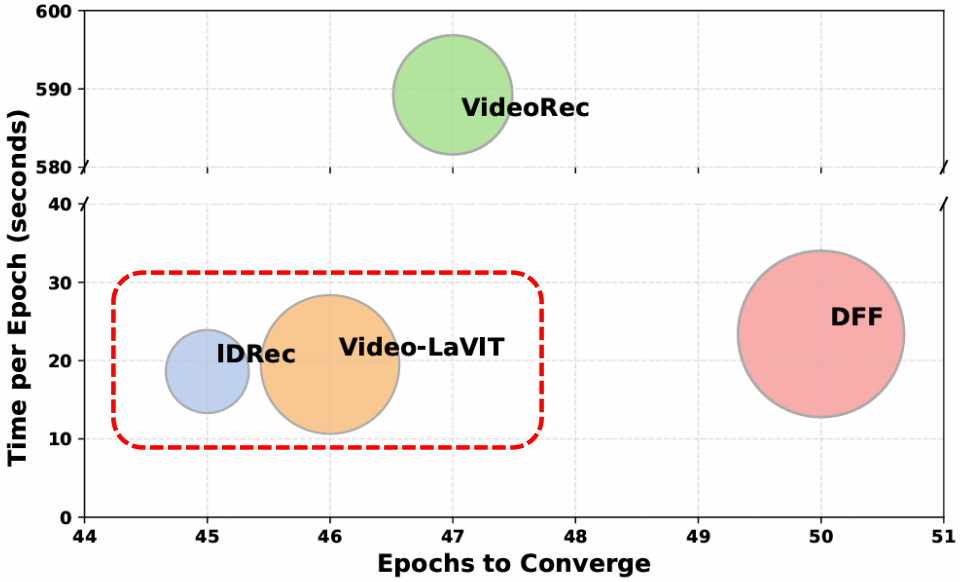
The results confirm that adaptive fusion is effective and transferable for frozen LVLM-based recommendation.

Sun, Ma, et al. Frozen LVLMs for Micro-Video Recommendation: A Systematic Study of Feature Extraction and Fusion. ICMR 2026 (Best Paper Award).

© Copyright Singapore Management University. All Rights Reserved.

DFF: Training and Inference Efficiency

EFFICIENCY



Efficiency Takeaways

- DFF only introduces modest overhead over IDRec and Video-LaVIT.
- DFF is much more efficient than VideoRec:
  25.1× faster training
  25.4× faster inference
- DFF runs on a single RTX 4090, while VideoRec requires two A800 GPUs.

© Copyright Singapore Management University. All Rights Reserved.

PART II

From Items to Bundles

Combinatorial explosion: generate many items at once, not one by one

DDBC (ICLR 2026)

© Copyright Singapore Management University. All Rights Reserved.

From Single Items to Bundles

Example bundles in various applications

Software



Electronics



Fashion



Meal



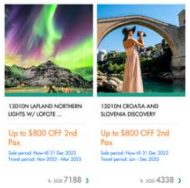
Furniture



Cosmetics



Travel



Video Game



Online Game Shopping





Music Playlist



Some synonyms: next **basket** recommendation, **item set** recommendation

© Copyright Singapore Management University. All Rights Reserved.

Bundle Completion: Combinatorial Explosion

Given a partial bundle, select the missing items from the whole catalog, as a set



Observed | **One compatible completion**

No order
A playlist or an outfit is a set

$k \leq 90$
items per Spotify playlist

$N = 254{,}155$
tracks in the catalog

About 10^{486} possible bundles

Tu, Ai, et al. Discrete Diffusion for Bundle Construction. ICLR 2026.

© Copyright Singapore Management University. All Rights Reserved.

DDBC: Why Not AR or One-Shot?

a) **AR-style** bundle construction inherits a left-to-right factorization,

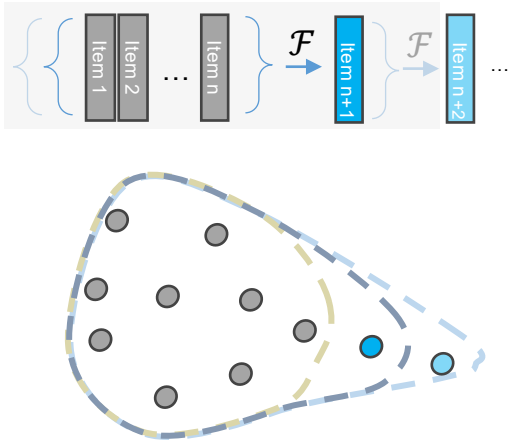
⚠ 1. **Overemphasis on order**: Sparse or even misleading supervision

⚠ 2. **Unidirectionally**: Bundles require modeling dense higher-order dependencies

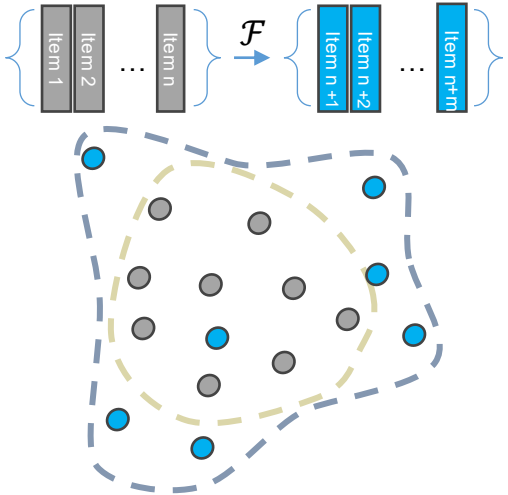
b) **One-Shot-style** adopts an over-simplified dependence factorization,

⚠ Omit relations **within the generated items**

(a) Next-item Prediction



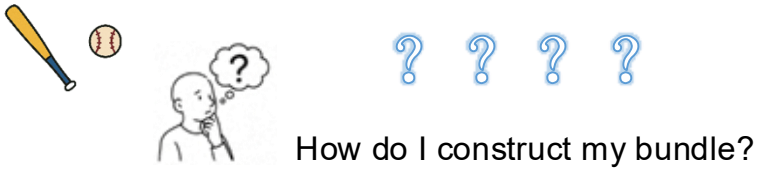
(b) One-Shot Completion



© Copyright Singapore Management University. All Rights Reserved.

DDBC: How Do People Build a Bundle?

①



How do I construct my bundle?

②



At least I need to update my shirt and shoes

③



This shoes look great!

④



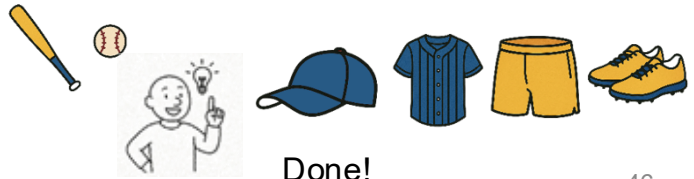
Let me think…
The shirt shall be…

⑤



The idea is clear!

⑥



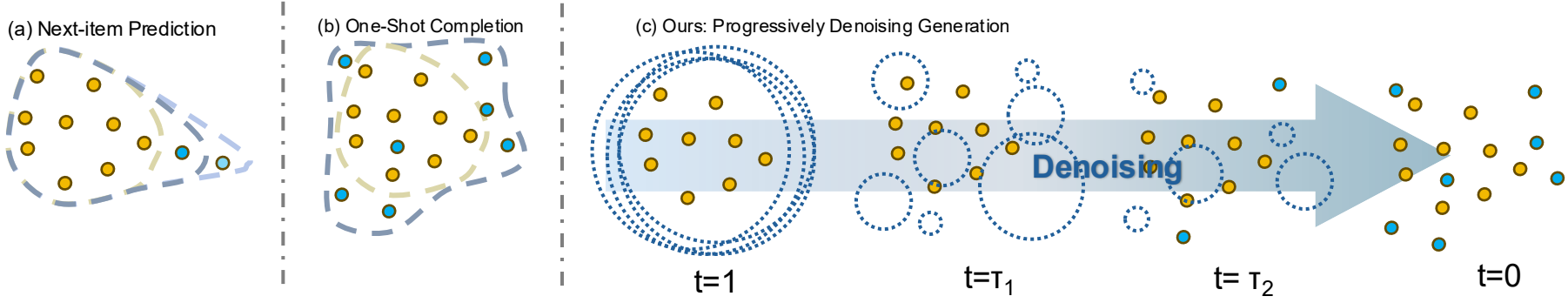
Done!

© Copyright Singapore Management University. All Rights Reserved.

DDBC: Bundle Construction as Denoising

- We aim for:
  - Need to be order invariance $P(N,k) \longrightarrow C(N,k)$
  - A better modeling of the intra-bundle relations

Diffusion Model could be an option



© Copyright Singapore Management University. All Rights Reserved.

DDBC: Compact Item Tokens via RVQ

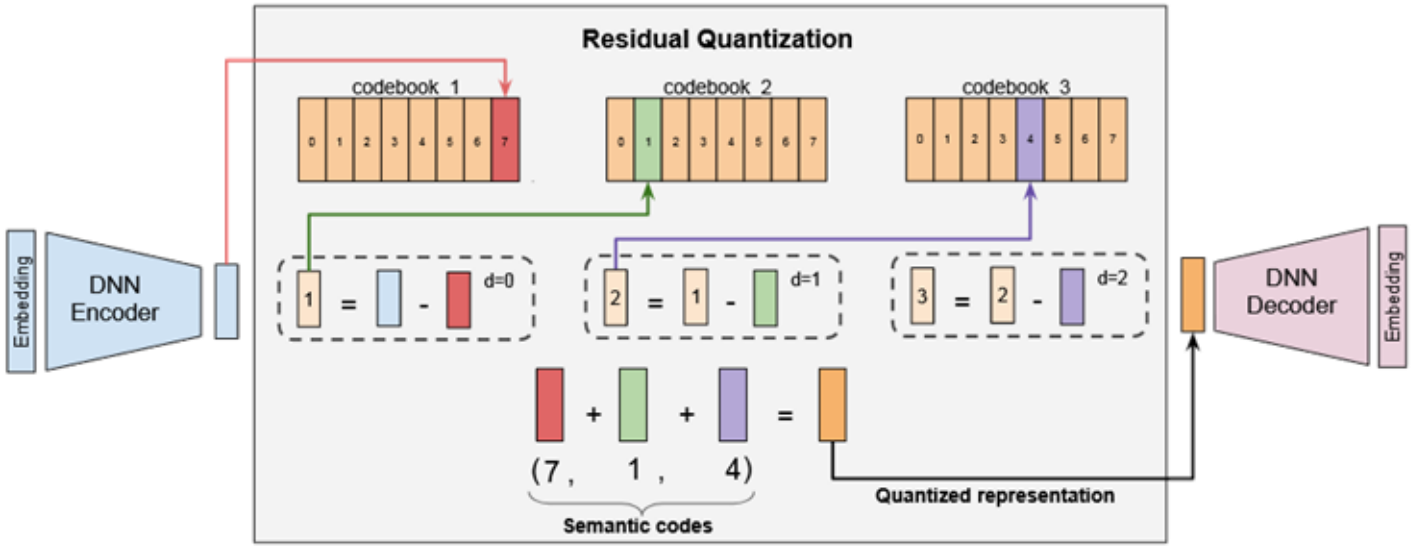
EFFICIENCY

Core Idea:

Use **Residual Vector Quantization** to quantize item embeddings into multiple discrete tokes.

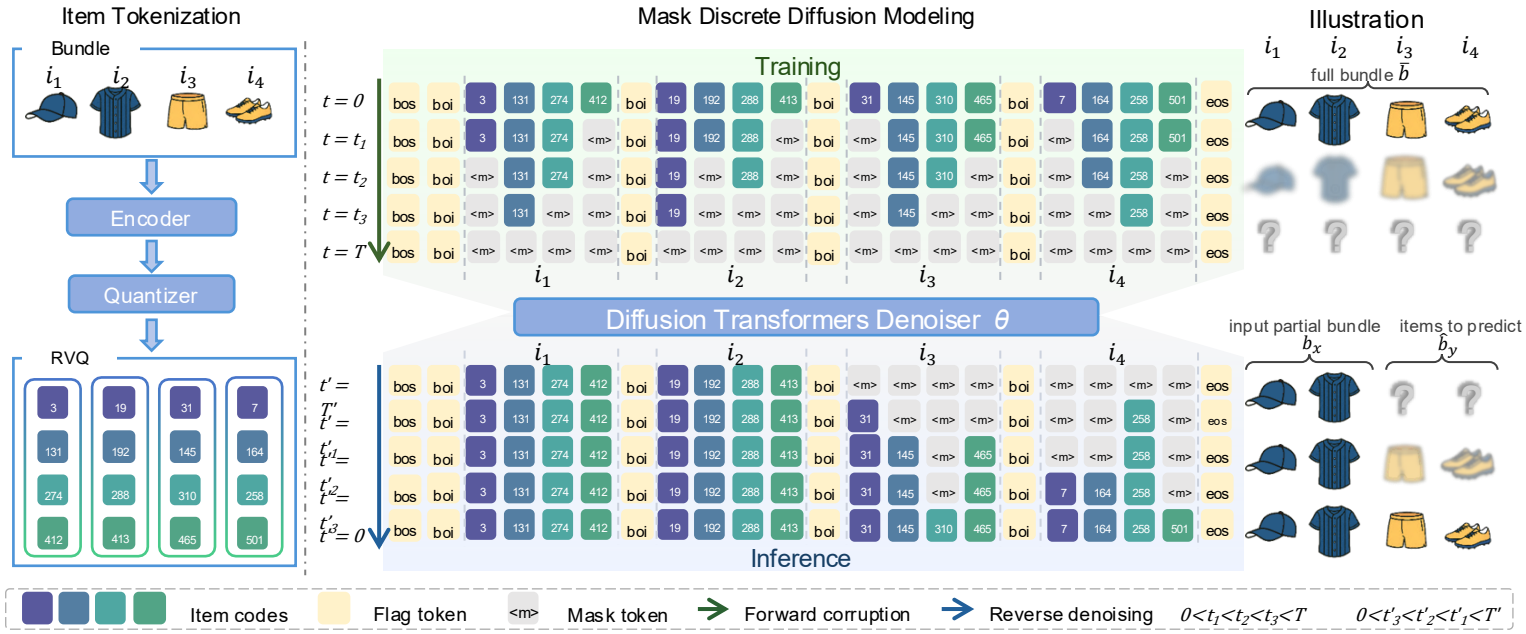
Factorizes a large space into smaller codebooks

Reduces per-codebook vocabulary size: $N = 254,151$ **v.s.** $V = 128 \times 4$



© Copyright Singapore Management University. All Rights Reserved.

DDBC: Framework (ICLR 2026)



© Copyright Singapore Management University. All Rights Reserved.

DDBC: Results

Table 1: Overall performance comparison between our DDBC and baselines. "%Improv." denotes the relative improvement over the strongest baseline. Best in **bold**, second best underlined.

Model (A/beam)	Spotify$_{k=30}$			Spotify$_{k=60}$			Spotify$_{k=90}$			POG$_{k=4}$		
	F1 ↑	Jacc ↑	OAS ↑	F1 ↑	Jacc ↑	OAS ↑	F1 ↑	Jacc ↑	OAS ↑	F1 ↑	Jacc ↑	OAS ↑
CLHE	0.071	0.039	0.373	0.100	0.054	0.446	0.119	0.065	0.486	0.140	0.096	0.446
Bi-LSTM	0.124	0.071	0.489	0.062	0.034	0.430	0.047	0.025	0.426	0.035	0.024	0.390
SASRec	0.070	0.043	0.318	0.089	0.054	0.310	0.050	0.029	0.285	0.169	0.114	0.468
TIGER	0.093	0.053	0.329	0.129	0.076	0.413	0.123	0.070	0.480	**0.213**	**0.157**	**0.546**
BundleNAT	0.153	0.090	0.454	0.101	0.056	0.438	0.095	0.052	0.446	0.145	0.097	0.462
BundleMLLM	0.046	0.024	0.296	0.045	0.024	0.324	0.052	0.027	0.355	0.070	0.047	0.322
DDBC	**0.282**	**0.178**	**0.618**	**0.296**	**0.185**	**0.668**	**0.287**	**0.177**	**0.684**	0.139	0.098	0.526
%Improv. +	84.3%	97.8%	26.4%	129.5%	143.4%	49.8%	133.3%	152.9%	40.7%	–	–	–

DDBC performance ↑ as bundle length ↑

→ DDBC effectively captures the higher-order intra-bundle item relations, particularly for long-sequence bundles with rich structural dependencies.

© Copyright Singapore Management University. All Rights Reserved.

DDBC: Efficiency

EFFICIENCY

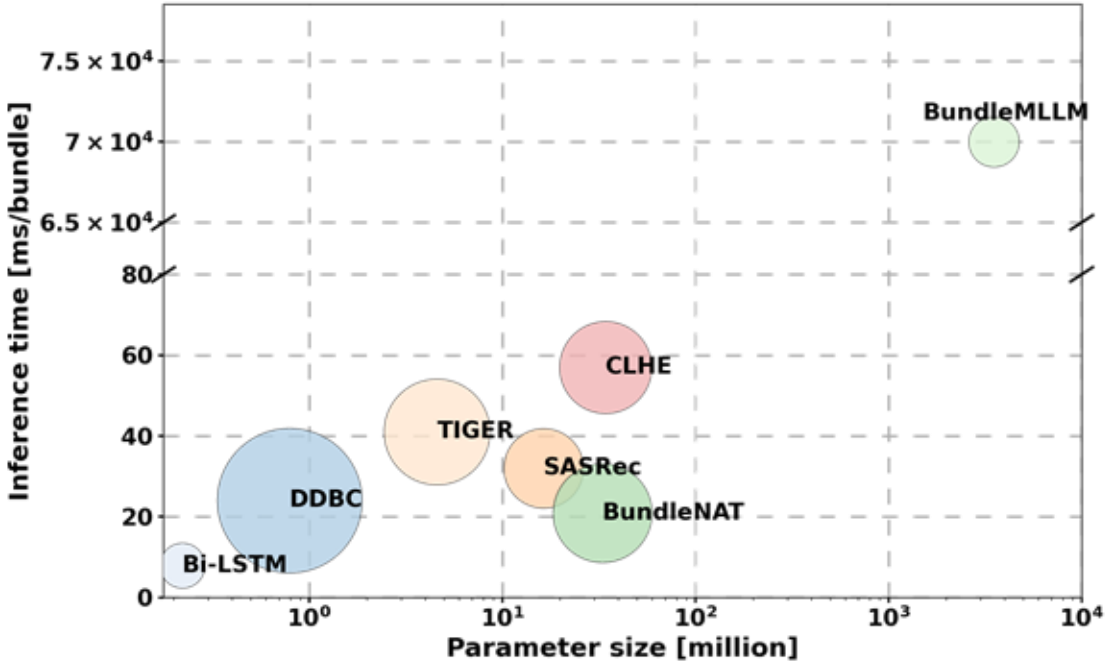


Illustration of the model efficiency comparison.
bubble radius corresponds to overall performance (larger is better).

© Copyright Singapore Management University. All Rights Reserved.

PART III

Personalized Generation

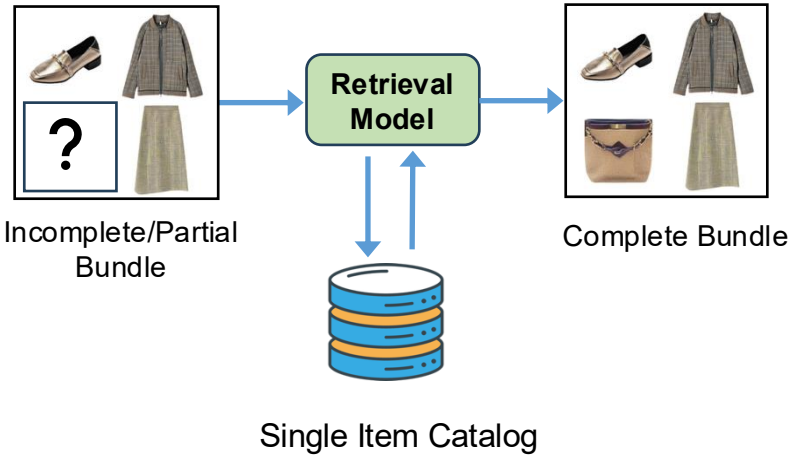
Generate content: create new items that users want

DiFashion (SIGIR 2024) · FashionDPO (SIGIR 2025) · DualFashion (SIGIR 2026)
R2G-Map (ongoing work)

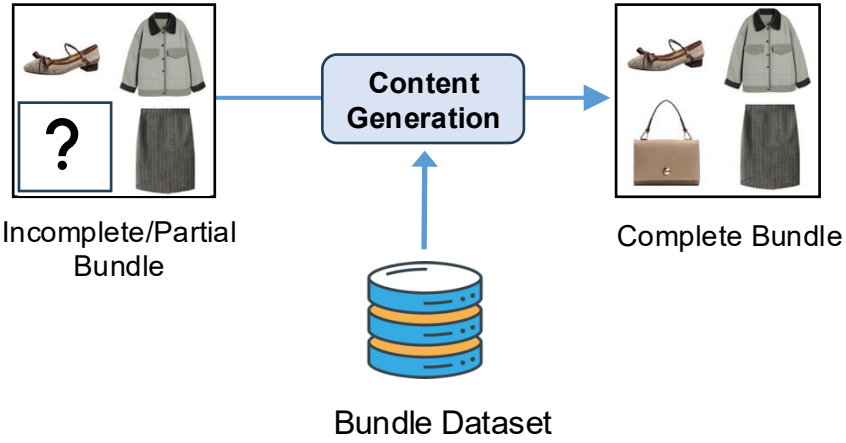
© Copyright Singapore Management University. All Rights Reserved.

Beyond Retrieval: Generate New Items

Paradigm 1: Retrieval



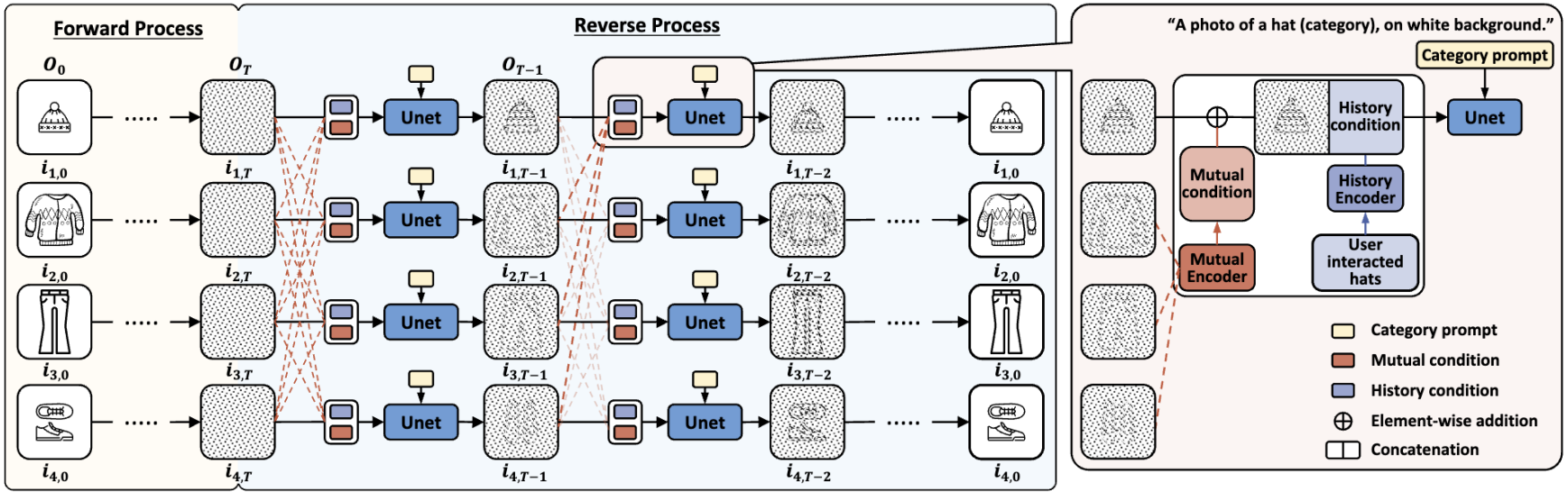
Paradigm 2: Content Generation



© Copyright Singapore Management University. All Rights Reserved.

DiFashion: Generative Outfits (SIGIR'24)

DiFashion: Diffusion Models for Generative Outfit Recommendation



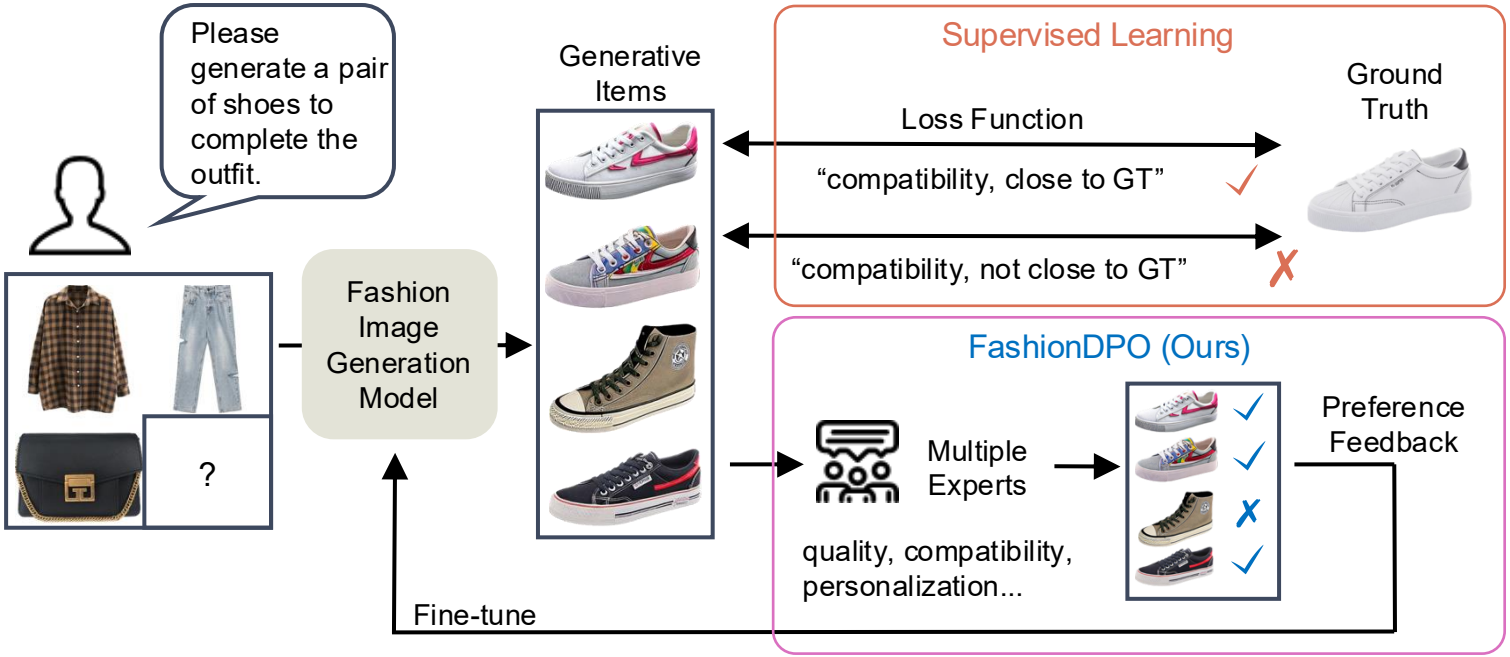
1) Use diffusion model to directly generate fashion item images
2) Condition the generation with both user's historical interaction (personalization) and other items within the bundle (compatibility)

Diffusion Models for Generative Outfit Recommendation. Yiyan Xu et al. SIGIR 2024.

© Copyright Singapore Management University. All Rights Reserved.

FashionDPO: Expert Feedback (SIGIR'25)

FashionDPO: fine-tune fashion outfit generation model using direct preference optimization

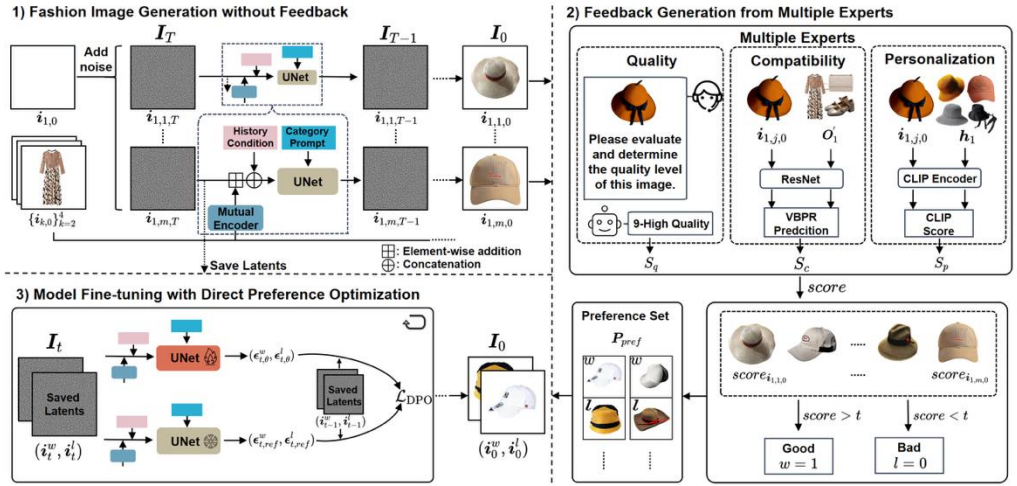


FashionDPO: Fine-tune Fashion Outfit Generation Model using Direct Preference Optimization. Mingzhe Yu et al. SIGIR 2025.

© Copyright Singapore Management University. All Rights Reserved.

DualFashion Motivation: Slow Rollouts

EFFICIENCY



FashionDPO pipeline: roll out, score with experts, DPO fine-tune, repeat

Every training round:

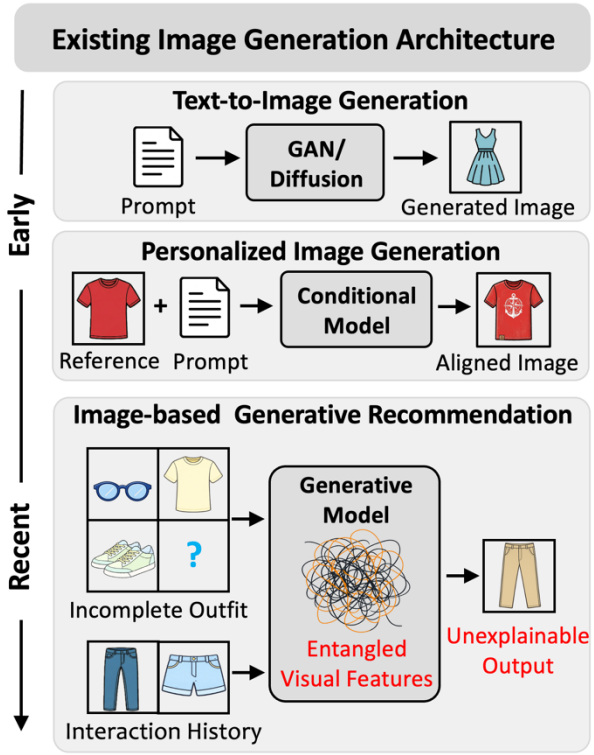
- Roll out several images per outfit
- Each one is a full diffusion sampling
- Score them with three experts
- Build preference pairs, then DPO



Training time, 15k steps on L40S

© Copyright Singapore Management University. All Rights Reserved.

DualFashion: Limits of Image-Only Models



Limitations:

1. User behavior modeling remains insufficient.

- interacted item images as unimodal inputs
- user–item interactions are inherently sparse
- visual data often contain redundant or preference-irrelevant details

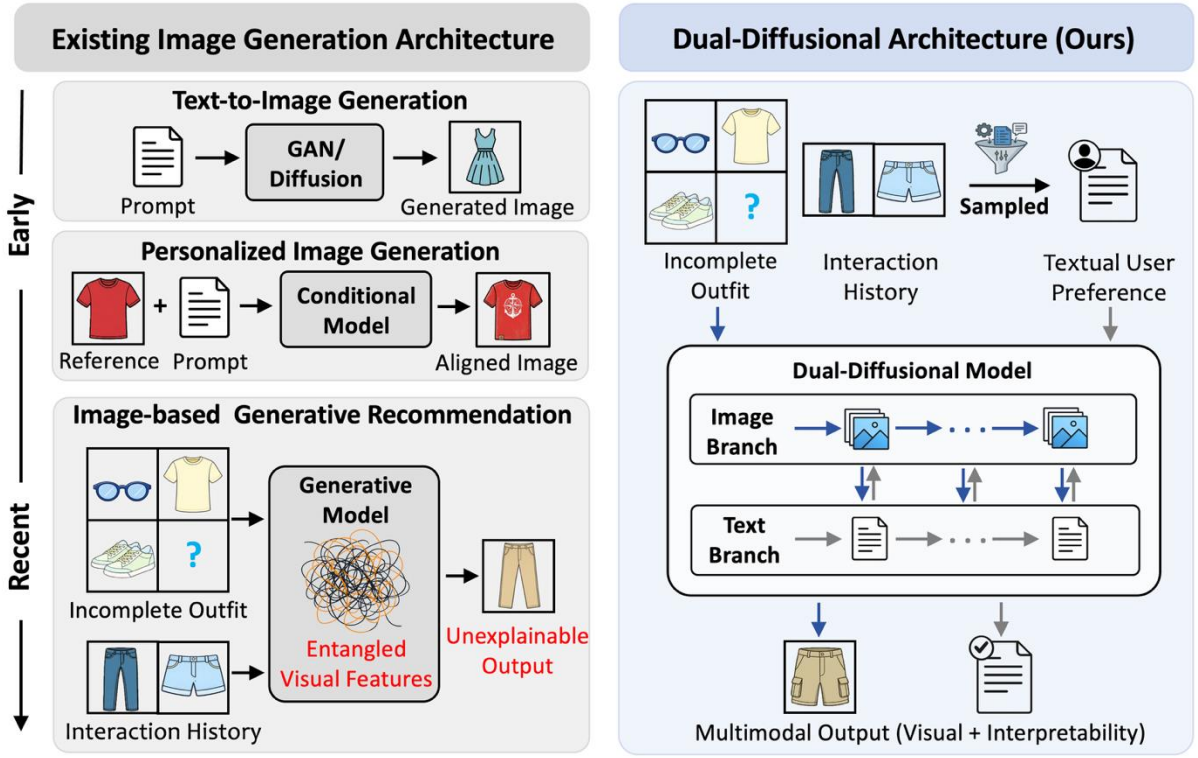
2. Explainability is largely absent.

- generation models treat image generation as the sole output
- solutions are typically add-on patches to image-centric backbones

Common root cause:
the restrictive image-centric architectural design

© Copyright Singapore Management University. All Rights Reserved.

DualFashion: Joint Image–Text Diffusion



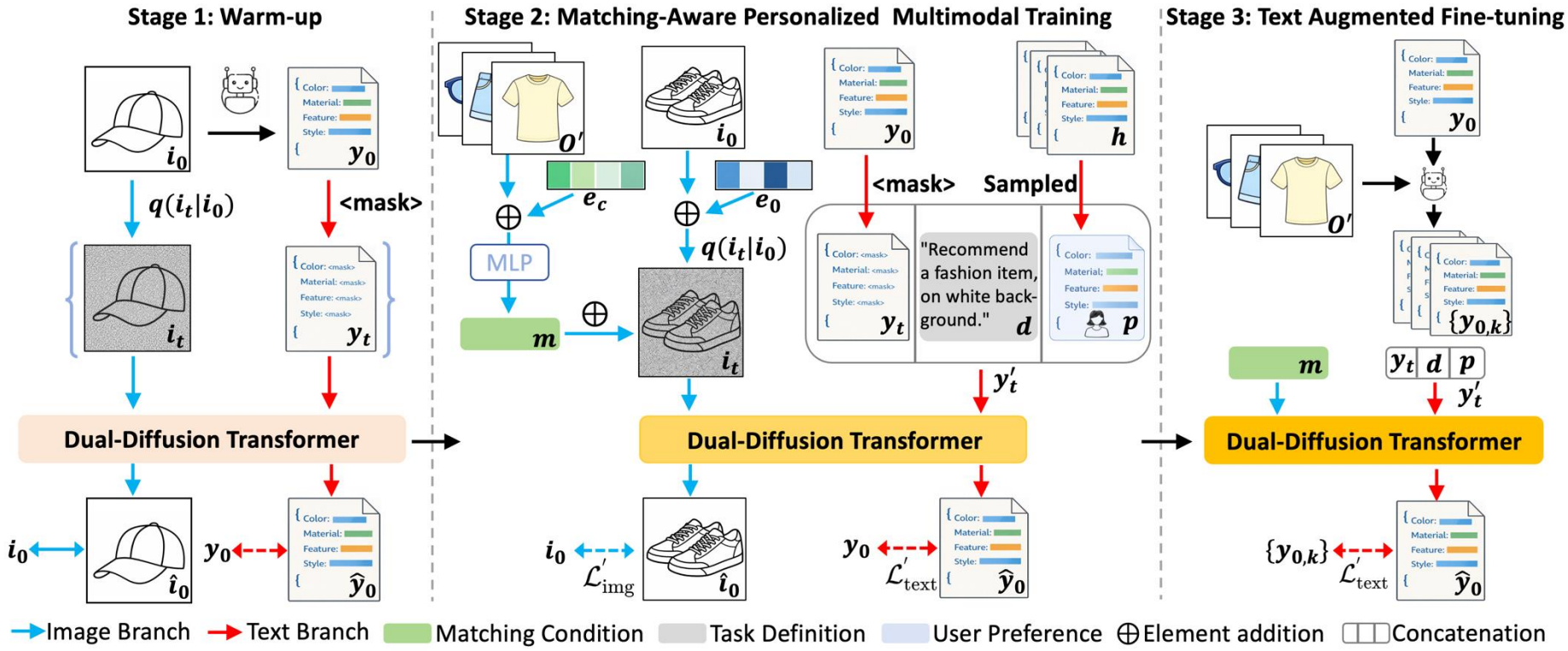
Multimodal architectures

- Sample textual user preference
- Support image–text inputs
- Image Branch
- Text Branch
- Jointly generate visual and textual outputs

© Copyright Singapore Management University. All Rights Reserved.

DualFashion: Text-Augmented Fine-Tuning

EFFICIENCY



© Copyright Singapore Management University. All Rights Reserved.

DualFashion: Overall Performance

Overall Performance Comparison

Dataset	iFashion									Polyvore-U								
Task	PFITB					GOR				PFITB					GOR			
Evaluation metric	IS	IS-acc	Comp.	Per.	Div.	IS	IS-acc	Per.	Div.	IS	IS-acc	Comp.	Per.	Div.	IS	IS-acc	Per.	Div.
SD-v1.5 [32]*	22.54	0.76	0.08	46.31	0.56	23.20	0.77	46.45	0.57	17.10	0.73	0.70	51.05	0.55	16.95	0.73	50.99	0.52
SD-v2 [32]*	21.66	0.71	0.04	46.60	**0.62**	22.19	0.74	46.60	**0.59**	14.83	0.68	0.60	51.29	**0.60**	14.88	0.67	51.23	**0.65**
SD-v1.5 [32]	26.76	0.83	0.46	53.16	0.50	26.90	0.84	53.24	0.49	17.12	0.72	0.75	58.20	0.50	17.24	0.72	58.16	0.47
SD-v2 [32]	25.85	0.80	0.39	52.99	0.47	25.82	0.82	53.06	0.47	15.59	0.67	0.71	58.79	0.46	16.33	0.70	58.91	0.44
SD-naive [32]	25.45	0.80	0.36	52.95	0.39	25.43	0.81	52.95	0.40	15.45	0.66	0.73	59.24	0.41	15.48	0.67	59.12	0.40
ControlNet [48]	27.76	0.81	0.16	49.90	0.41	28.49	0.82	49.91	0.43	18.93	0.77	0.73	55.44	0.42	19.21	0.77	55.40	0.41
DiFashion [42]	29.99	0.90	0.58	55.86	0.33	30.04	0.90	55.54	0.33	19.67	0.84	0.80	61.44	0.37	18.95	0.83	61.16	0.36
FashionDPO [47]	**33.80**	**0.91**	**0.74**	60.39	0.36	**32.37**	**0.91**	59.98	0.35	24.14	**0.89**	**0.83**	64.67	0.38	**24.93**	0.87	64.79	0.36
DualFashion	31.08	0.88	0.69	**65.17**	0.38	31.53	0.89	**64.05**	0.39	**25.30**	**0.89**	0.81	**67.40**	0.42	24.85	**0.88**	**67.13**	0.40

- Slightly lower scores than FashionDPO in terms of IS, IS-acc, and Comp
- Significant improvement in the Personalization score
- DualFashion has higher Diversity compared with FashionDPO; Stable Diffusion models exhibit the highest Diversity scores, but stems from randomness and lack of constraints

© Copyright Singapore Management University. All Rights Reserved.

DualFashion: Qualitative Results

Qualitative Results - Generative Outfit Recommendation (GOR)



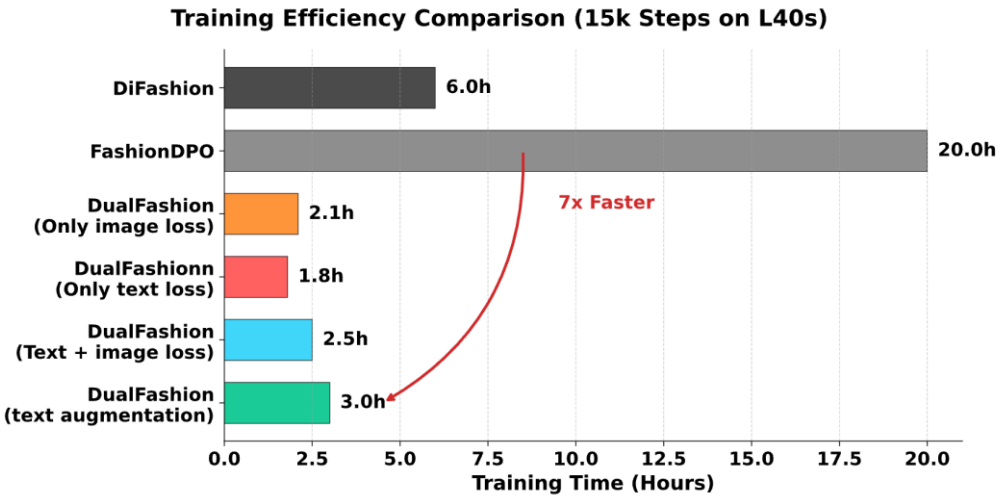
- **Quality**: all models generate visually plausible items
- **Compatibility**: Baselines generate items that match individual pieces but lack global coherence
- **Personalization**: DualFashion better reflects user preferences observed from interacted items

© Copyright Singapore Management University. All Rights Reserved.

DualFashion: ~7× Faster Training

EFFICIENCY

Time cost analysis



- DualFashion is approximately **7x faster** than FashionDPO, which significantly reduces the overall training time.
- predict only masked captions and optimizing the text loss incurs the lowest computational cost among all settings

© Copyright Singapore Management University. All Rights Reserved.

R2G-Map (Ongoing): Motivation

Generative retrieval

TIGER, DDBC

+ Learns user and bundle tastes
− Returns catalog items only

R2G-Map

Image generation

diffusion, visual tokens

+ Creates new, unseen content
− Blind to learned preferences

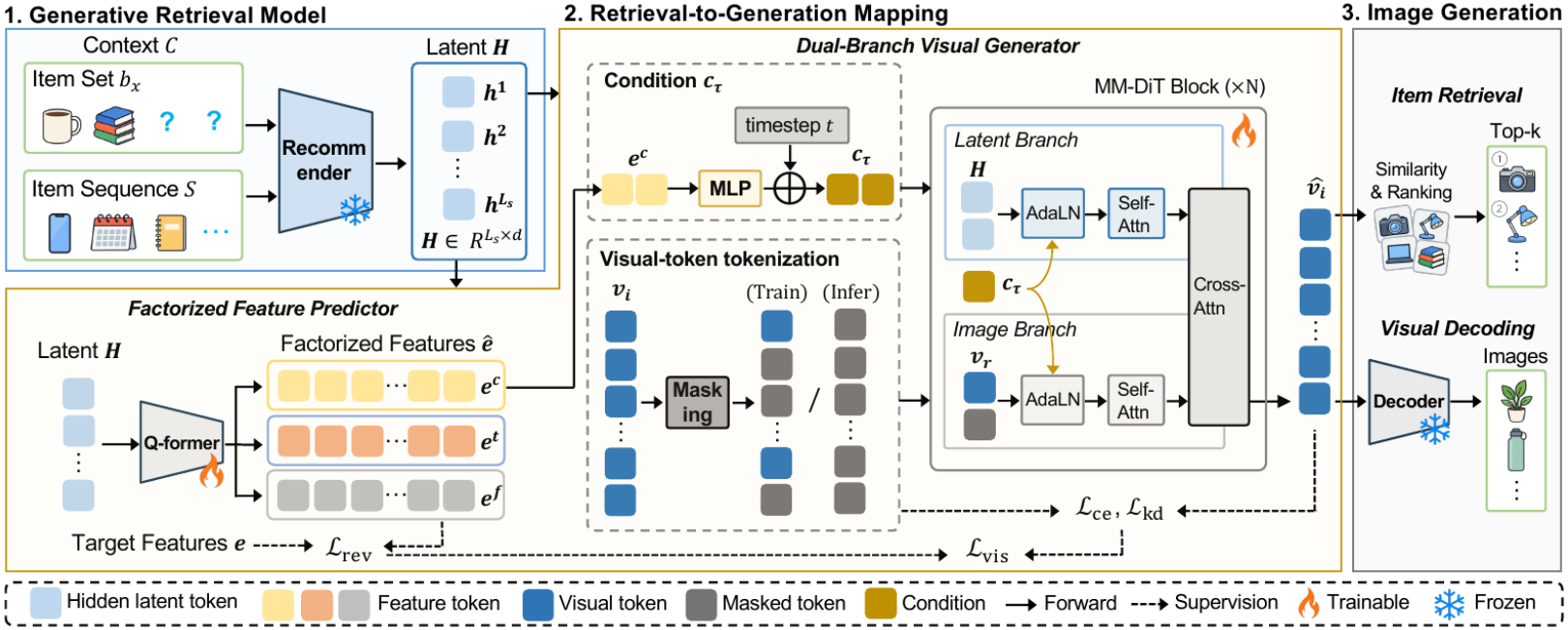
Question: can a generator reuse the preferences a recommender has already learned?

Efficiency: no new preference training for the generator; reuse a frozen recommender and a frozen decoder

ONGOING WORK

© Copyright Singapore Management University. All Rights Reserved.

R2G-Map (Ongoing): Method



1 **Frozen recommender**
hidden states before item selection

2 **Feature predictor**
Q-Former recovers content features

3 **Visual generator**
denoises masked visual tokens

ONGOING WORK

© Copyright Singapore Management University. All Rights Reserved.

R2G-Map (Ongoing): One Mapping, Two Outputs



Case: same outfit context; DiFashion, FashionDPO and R2G-Map on DDBC / TIGER

Output 1 · Rank catalog items

Score candidate images with the predicted visual-token distribution

Output 2 · Create new items

Sample visual tokens and decode them with a frozen decoder

Works on top of both DDBC and TIGER

Under review. Experiments use outfit context built from POG, without long-term user history; generated items are plausible completions, not exact copies of the target.

ONGOING WORK

© Copyright Singapore Management University. All Rights Reserved.

Takeaways: an Efficiency Lever at Each Stage

Stage	Bottleneck	Our lever	What it buys
Represent content	Large foundation models are slow	LoRA distillation (TPAD) → frozen LVLM + light fusion (DFF)	25× faster training and inference; one RTX 4090
Compose items	Combinatorial explosion of item sets	RVQ item tokens + masked diffusion (DDBC)	1.8–2.3× F1 on long playlists
Generate content	Visual rollouts for preference tuning	Offload rollouts to the text branch; dual-diffusion backbone	~7× faster than FashionDPO; better personalization

Disclaimer: in most of these works, efficiency was a welcome side effect, not the main contribution.

Next: make efficiency the primary goal: keep accuracy unchanged while cutting cost.

THANKS and Q&A

© Copyright Singapore Management University. All Rights Reserved.